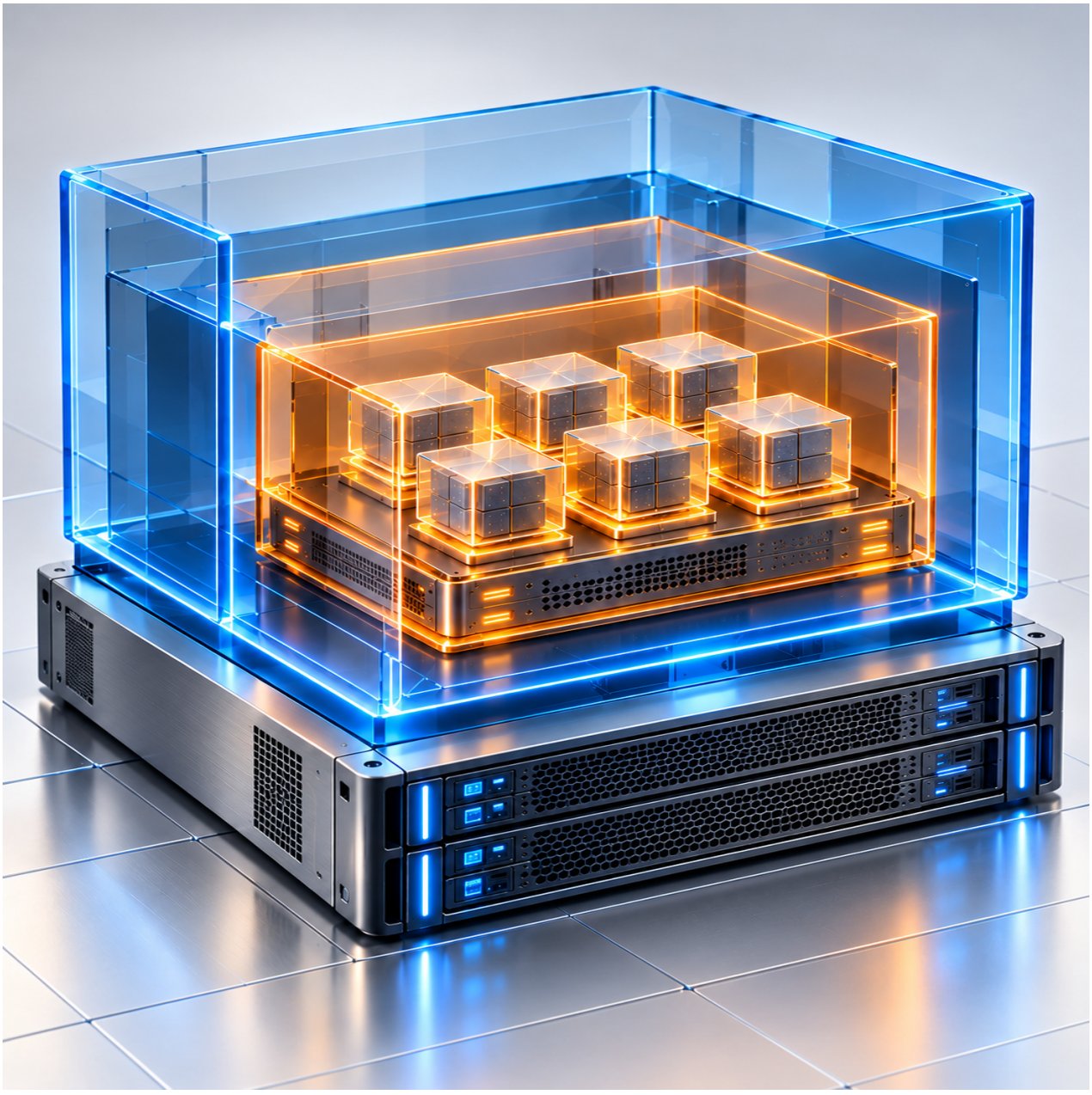
Kiformatou

DOCUMENTATION TECHNIQUE

Mise en place d’une infrastructure virtualisée

Procédures



Version : Windows 11 • **Date** : 29/10/2026 • **Auteur(s)** : Nathan Yang

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Présentation du projet

Cette documentation vise à accompagner le personnel informatique à mettre en place l'infrastructure de l'école Kiformatou. Le projet est limité par une configuration de 32 Go de RAM et 475 Go de stockage.

Architecture

L'infrastructure respectera l'architecture tant au niveau du plan d'adressage IP et organisation des appareils :

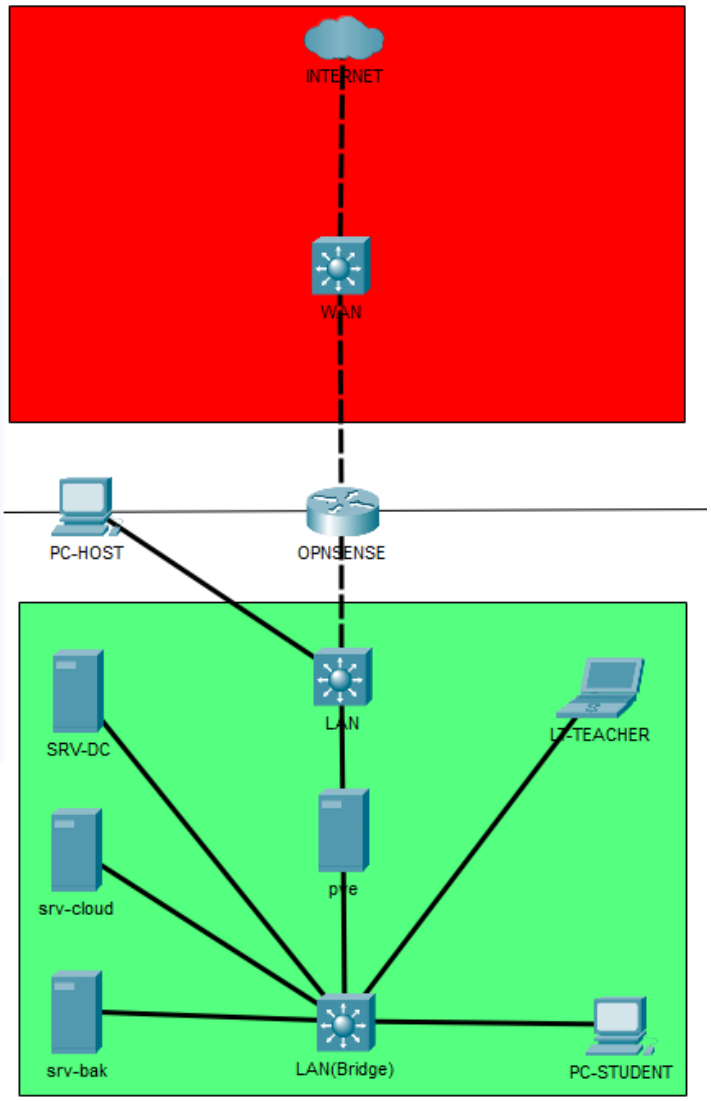
```
# LAN KIFORMATOU

## GENERAL
- @opnsense(WAN)     = DHCP
- @opnsense(LAN)     = 192.168.0.1/24
- @pve               = 192.168.0.2/24
- @PC-HOST           = 192.168.0.100/24

## VLAN 10 (SERVERS)
- hosts = 4
- @OPNSENSE(LAN)     = 192.168.10.1/29
- @SRV-DC            = 192.168.10.2/29
- @srv-cloud         = 192.168.10.3/29
- @srv-bak           = 192.168.10.4/29

## VLAN 20 (TEACHERS)
- hosts = 4
- @opnsense(LAN.20)  = 192.168.20.1/29
- @LT-TEACHER        = DHCP(192.168.20.2/29 - 192.168.20.5/29)

## VLAN 30 (STUDENTS)
- hosts = 96
- @pnsense(LAN.30)   = 192.168.30.1/25
- @PC-STUDENT        = DHCP(192.168.30.2/25 - 192.168.30.97/25)
```



Préparation de Hyper-V

Nous allons utiliser Hyper-V, un hyperviseur de type 1 pour simuler l'infrastructure de Kiformatou. Nous allons l'installer et le configurer afin de s'assurer que les VM puissent communiquer entre eux.

1. Activer Hyper-V

Sur Windows 10/11 Pro ou Entreprise, ouvrez PowerShell en administrateur, puis exécutez cette commande pour activer l'ensemble Hyper-V, y compris les outils de gestion et le module PowerShell :

```
Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V-All -All -NoRestart
```

Après l'installation, redémarrez :

```
Restart-Computer
```

Après le redémarrage, vérifiez les composants :

```
Get-WindowsOptionalFeature -Online |
    Where-Object FeatureName -Like "*Hyper-V*" |
    Format-Table FeatureName, State
```

Assurez-vous d'avoir l'affichage suivant où toutes les fonctionalités sont à **Enabled**:

```
FeatureName                                 State
-----------                                 -----
Microsoft-Hyper-V-All                     Enabled
Microsoft-Hyper-V                         Enabled
Microsoft-Hyper-V-Tools-All               Enabled
Microsoft-Hyper-V-Management-PowerShell   Enabled
Microsoft-Hyper-V-Hypervisor              Enabled
Microsoft-Hyper-V-Services                Enabled
Microsoft-Hyper-V-Management-Clients      Enabled
```

3. Créer les vNICS

Nous voulons créer un switch interne pour le LAN Kiformatou et un switch externe pour l'accès Internet. Exécutez les commandes suivantes pour les créer :

S'il n'y a pas de doublons, exécutez ces commandes pour créer les vSwitches de l'infrastructure sur l'hôte :

```
New-VMSwitch `
    -Name "KIFORMATOU" `
    -SwitchType Internal `


Rename-VMSwitch -Name "KIFORMATOU" -NewName "KIFORMATOU"


New-VMSwitch `
    -Name "INTERNET" `
    -NetAdapterName "Ethernet" `
    -AllowManagementOS $true
```

Vérifez ensuite :

```
Get-VMSwitch
```

Vous devriez obtenir une affichage similaire :

```
Name           SwitchType NetAdapterInterfaceDescription
----           ---------- ------------------------------
Default Switch Internal
KIFORMATOU     Internal
INTERNET       External   Realtek Gaming 2.5GbE Family Controller
```

(WIP : Vous pouvez également copier le script du dépôt de travail pour cette section)

Création de la VM OPNsense (REVOIR)

La VM OPNsense fera office de routeur/pare-feu pour relier le LAN Kiformatou à Internet, la création se fera en PowerShell. N'oubliez pas de prendre l'ISO de OPNsense v26.7 à cette adresse https://opnsense.org/download/

1. Créer la VM

Vous pouvez exécuter le script suivant pour créer la VM en adaptant le chemin absolu :

```
# Configuration
$OPNsense = [PSCustomObject]@{
    Name          = "opnsense"
    Generation    = 2
    CPUCount      = 2
    VRAMBytes     = 4GB
    Path          = "V:\Kiformatou\opnsense"
    VHDPath       = "V:\Kiformatou\opnsense\opnsense.vhdx"
    VHDSizeBytes  = 16GB
    SwitchNameWAN = "INTERNET"
    SwitchNameLAN = "KIFORMATOU"
    ISOPath       = "V:\ISO\opnsense-26.7.iso"
}

# Create the VM directory
New-Item -ItemType Directory -Path $OPNsense.Path -Force | Out-Null

# Create a Generation 2 VM with 4 GB RAM and a 64 GB virtual disk
New-VM -Name $OPNsense.Name `
    -Generation $OPNsense.Generation `
    -MemoryStartupBytes $OPNsense.VRAMBytes `
    -Path $OPNsense.Path `
    -NewVHDPath $OPNsense.VHDPath `
    -NewVHDSizeBytes $OPNsense.VHDSizeBytes `
    -SwitchName $OPNsense.SwitchNameWAN

# Assign 2 virtual processors and disable secure boot
Set-VMProcessor -VMName $OPNsense.Name -Count $OPNsense.CPUCount
```

```
Set-VMFirmware -VMName $OPNsense.Name -EnableSecureBoot Off


# Attach the installation ISO and boot from it
$dvdDrive = Add-VMDvdDrive -VMName $OPNsense.Name -
Path $OPNsense.ISOPath -Passthru
Set-VMFirmware -VMName $OPNsense.Name -FirstBootDevice $dvdDrive


# Disable dynamic memory
Set-VMMemory -VMName $OPNsense.Name -DynamicMemoryEnabled $false -
StartupBytes @OPNsense.VRAMBytes


# Add a second vNIC for LAN
Add-VMNetworkAdapter -VMName $OPNsense.Name `
    -Name $OPNsense.SwitchNameLAN `
    -SwitchName $OPNsense.SwitchNameLAN

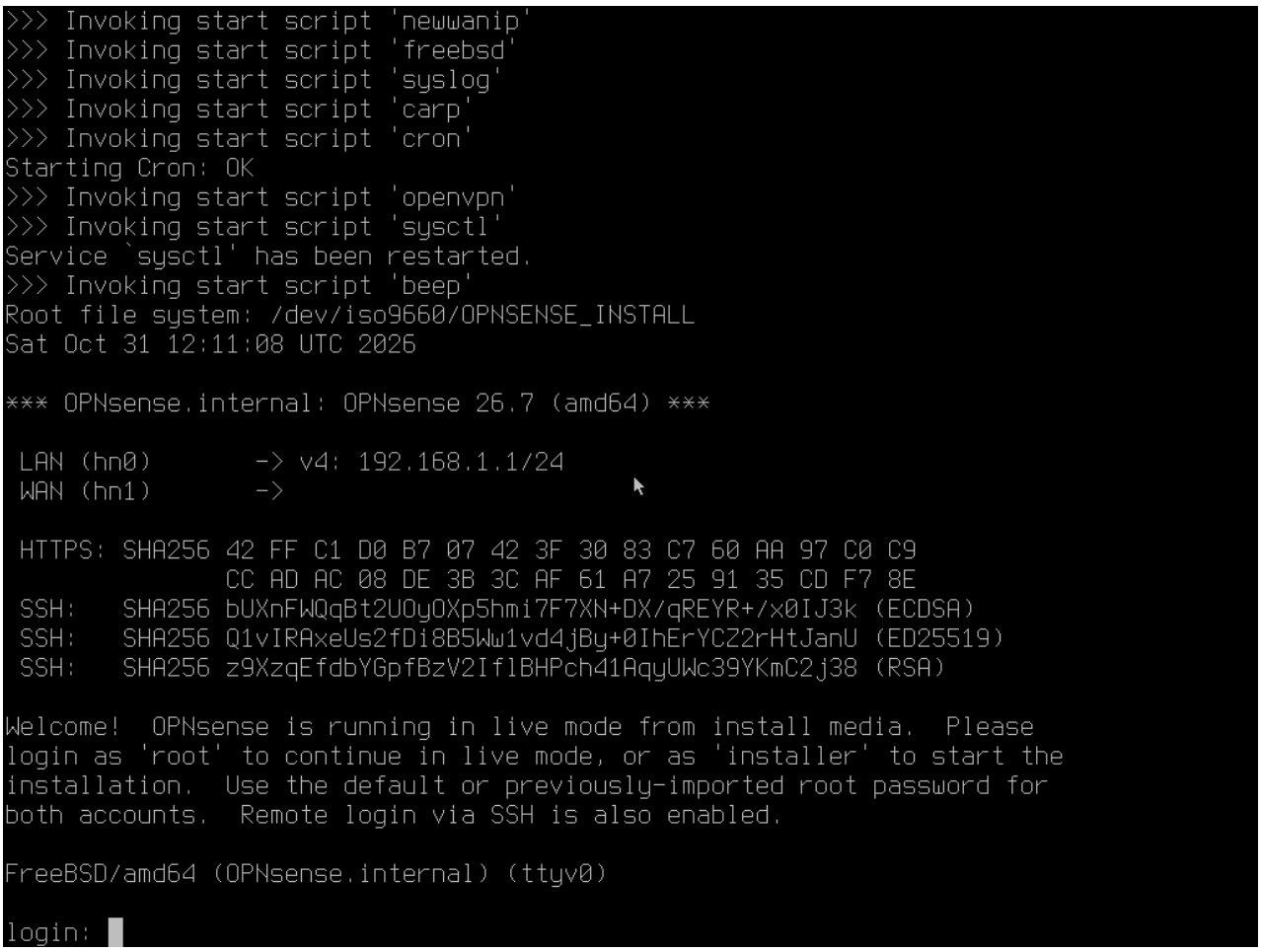

# Display the VM status
Get-VM -Name $OPNsense.Name
```

2. Installer la VM

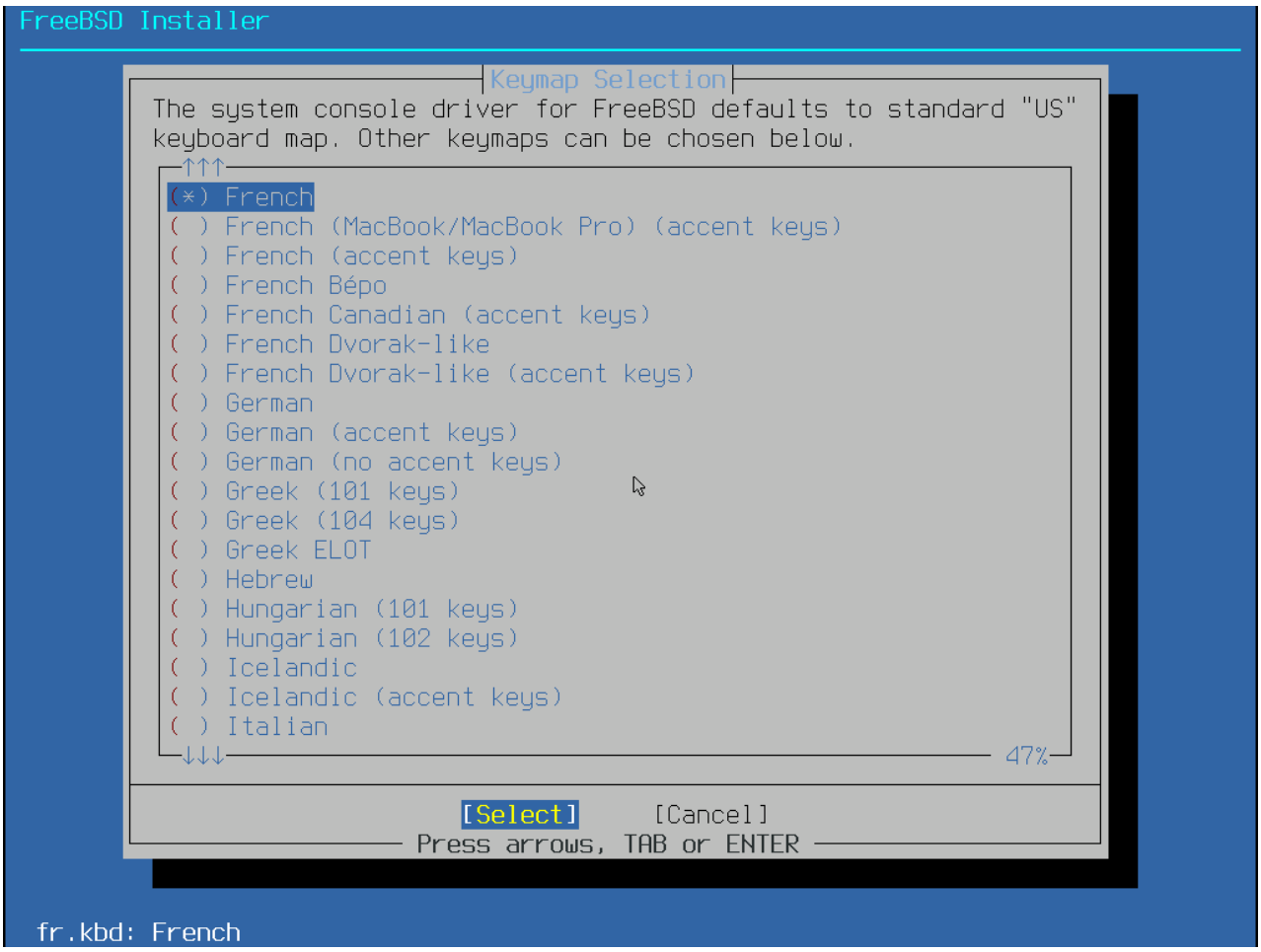
Démarrez la VM avec :

```
Get-VM
Start-VM -Name "opnsense"
```

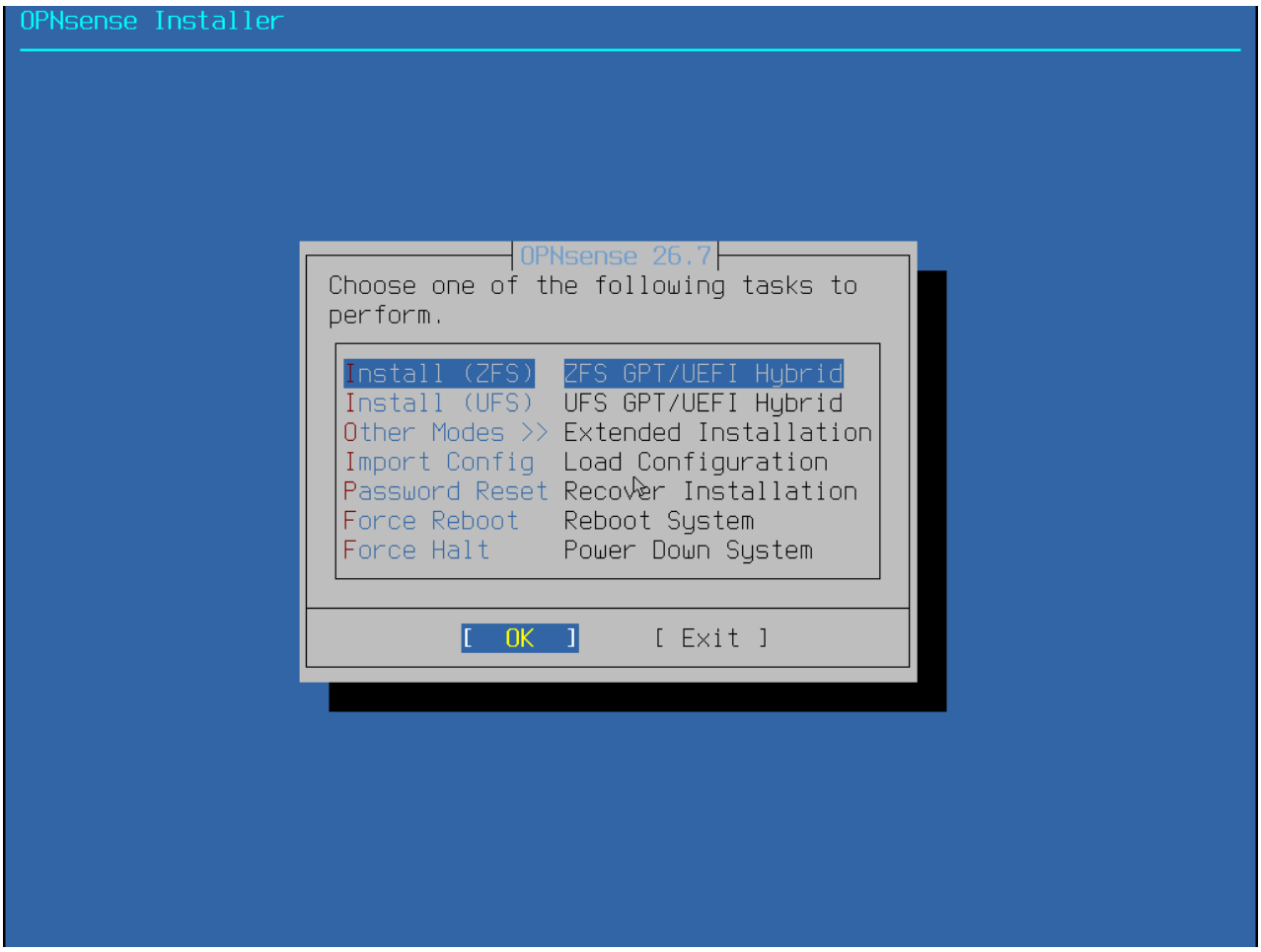
Puis attendez jusqu'à avoir cette affichage :



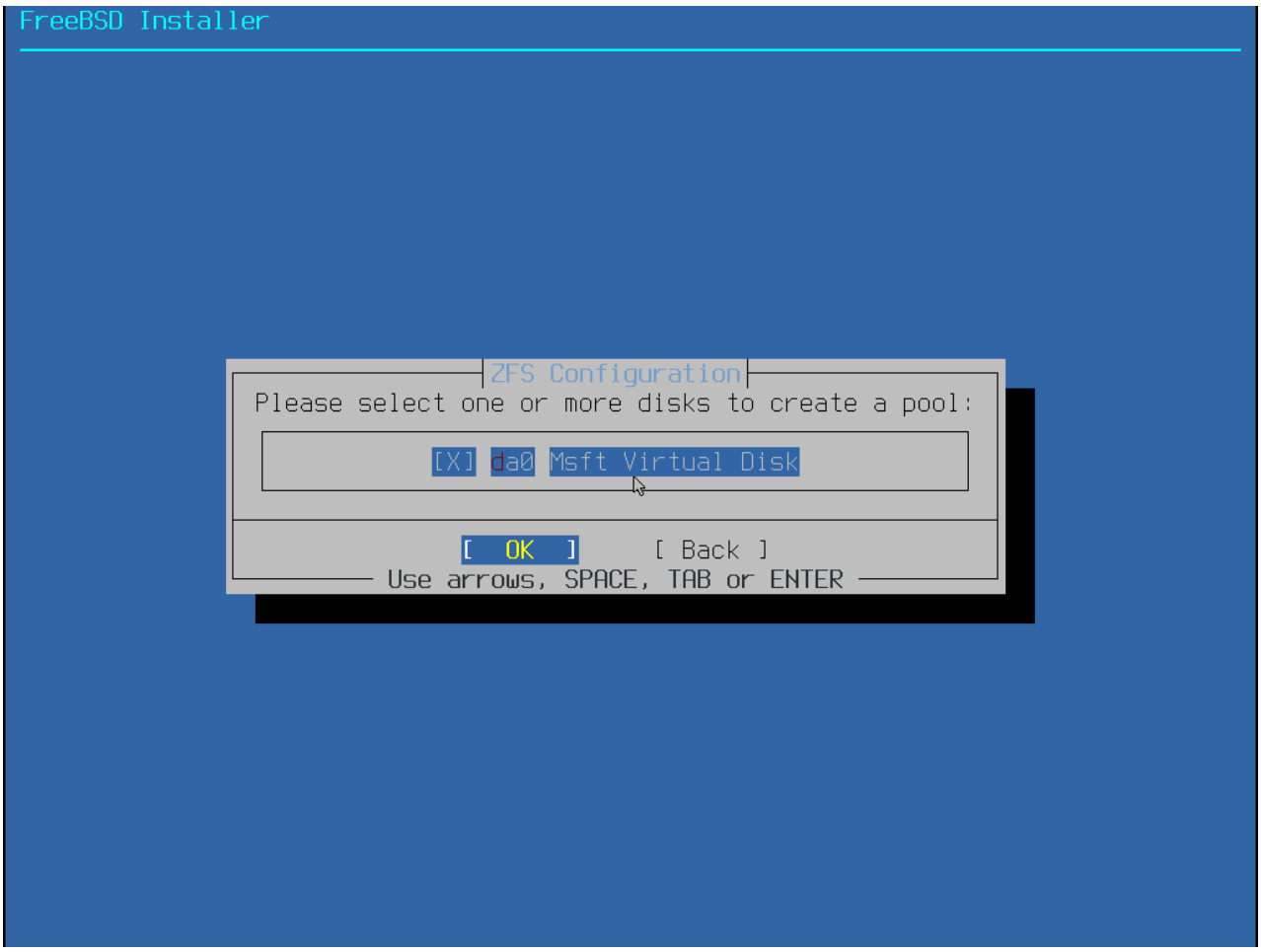
Rentrez l'identifiant **installer** et le mot de passe **opnsense** (ATTENTION : le clavier est en qwerty).



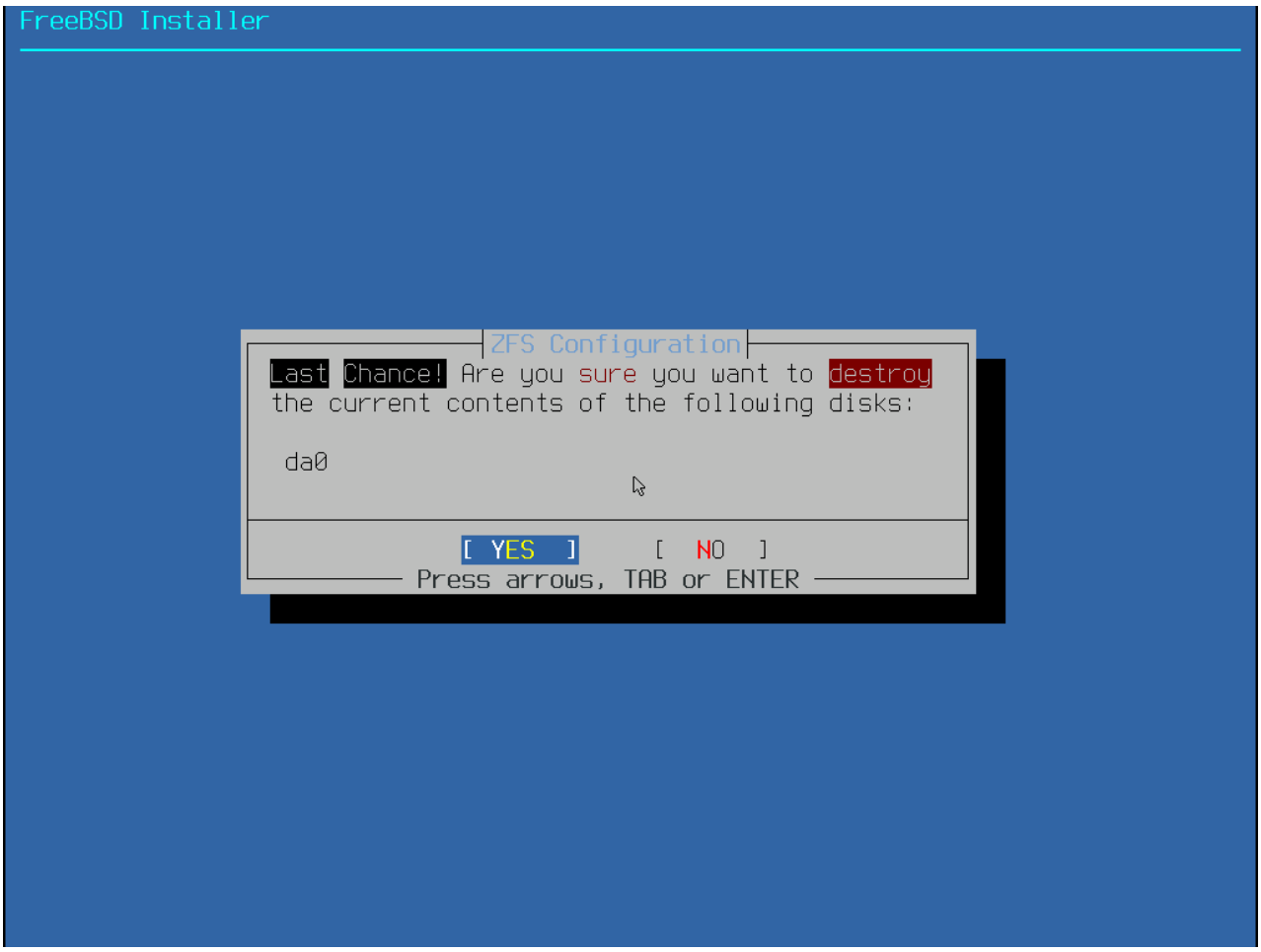
Choisissez **French**



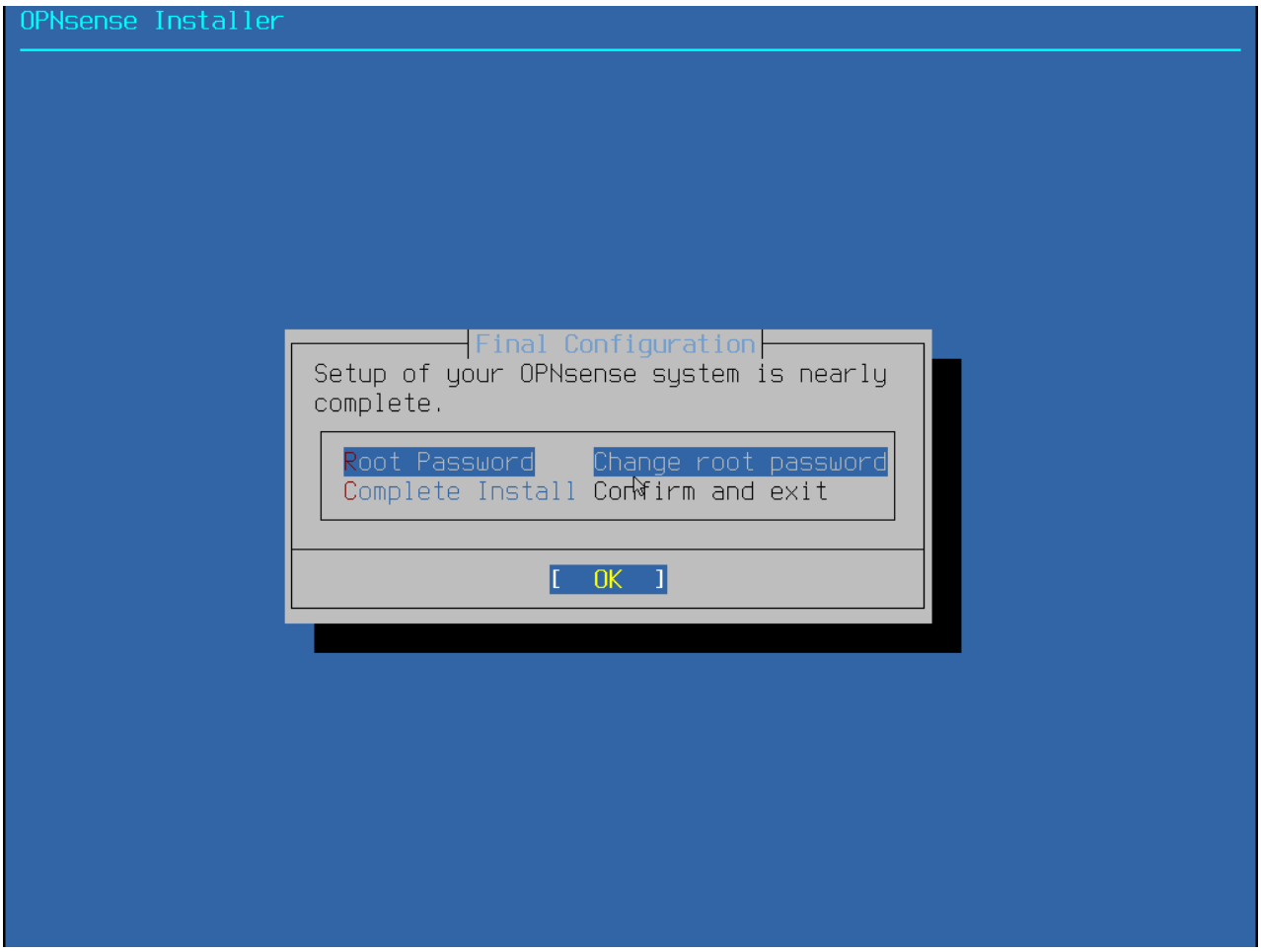
Continuez avec l'option par défaut **Install (ZFS)**

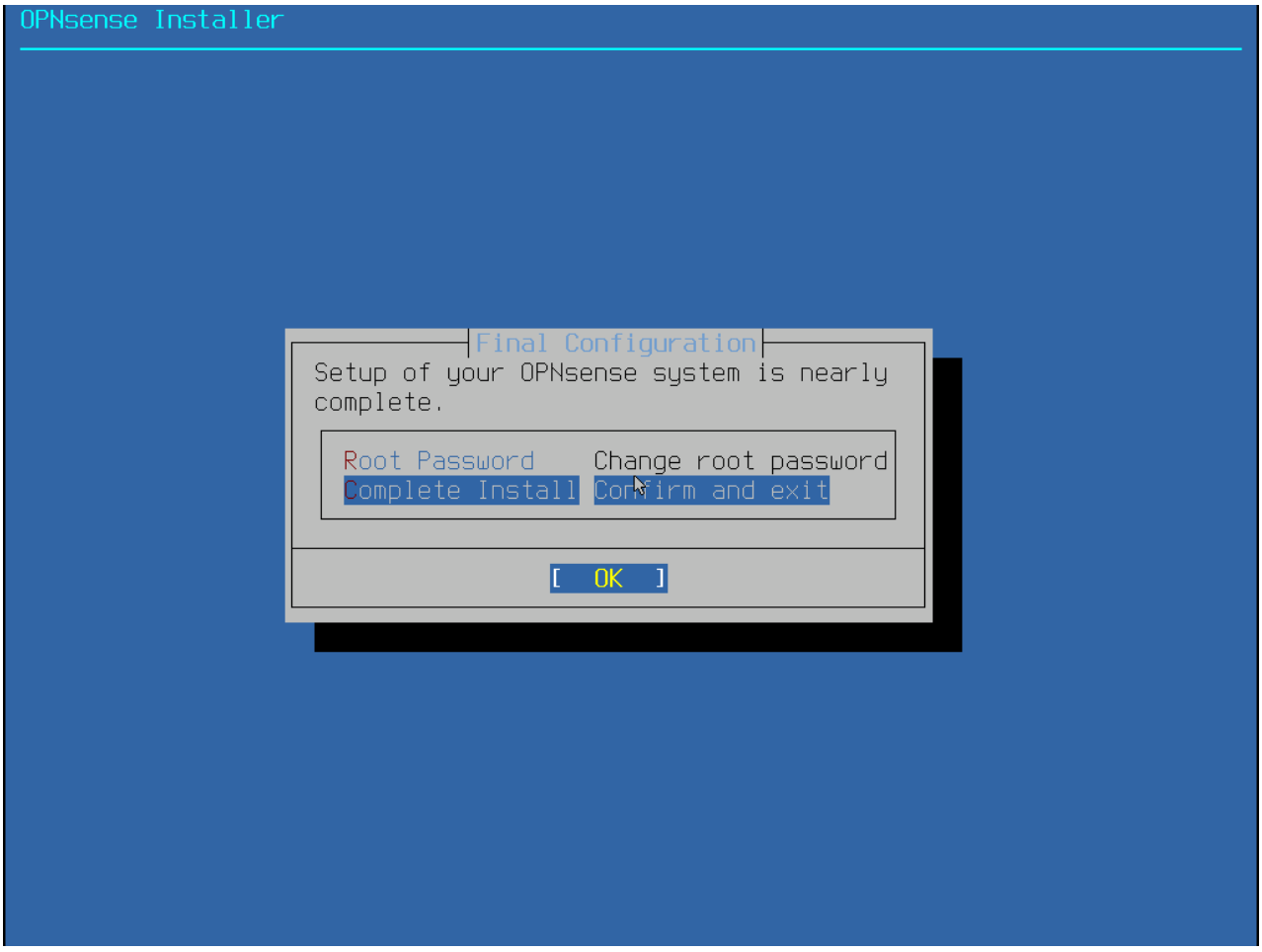


Sélectionnez l'unique disque et continuez

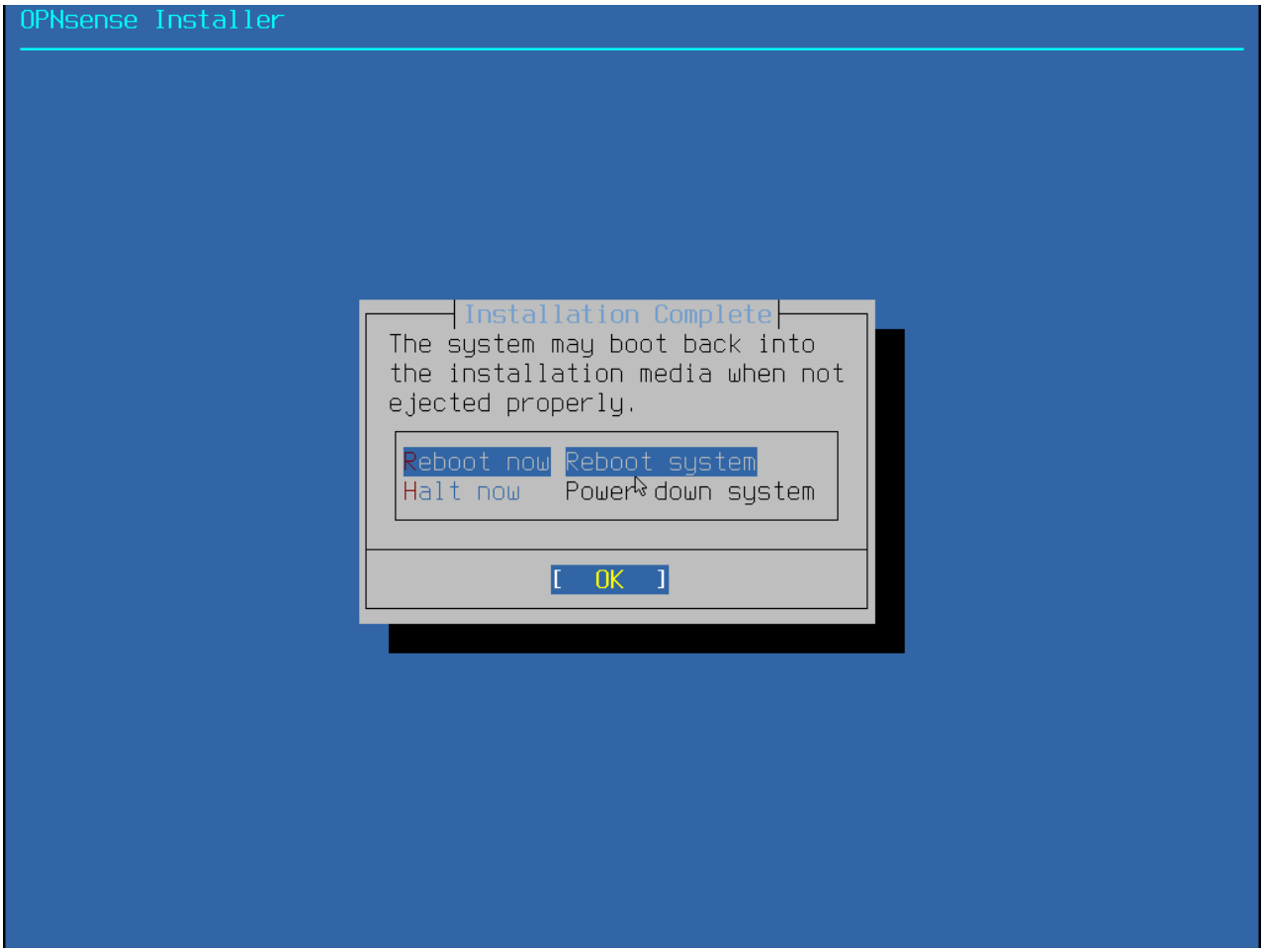

FreeBSD Installer
ZFS Configuration
Last Chance! Are you sure you want to destroy
the current contents of the following disks:
da0
[YES]
[NO]
Press arrows, TAB or ENTER

Confirmez


OPNsense Installer
Final Configuration
Setup of your OPNsense system is nearly complete.
Root Password Change root password
Complete Install Confirm and exit
[OK]


OPNsense Installer
Final Configuration
Setup of your OPNsense system is nearly complete.
Root Password Change root password
Complete Install Confirm and exit
[OK]

Changez le mot de passe du root et terminez l'installation



Redémarrez

3. Configurer la VM

Connectez-vous avec vos identifiants

3.1. Configurer les interfaces réseau

```
FreeBSD/amd64 (OPNsense.internal) (ttyv0)

login: root
Password:
-------------------------------------------------
|       Hello, this is OPNsense 26.7            |
|                                               |
|  Website:     https://opnsense.org/           |
|  Handbook:    https://docs.opnsense.org/      |
|  Forums:      https://forum.opnsense.org/     |
|  Code:        https://github.com/opnsense     |
|  Reddit:      https://reddit.com/r/opnsense   |
-------------------------------------------------

*** OPNsense.internal: OPNsense 26.7 (amd64) ***

 LAN (hn0)       -> v4: 192.168.1.1/24
 WAN (hn1)       ->

 HTTPS: SHA256 71 BC C5 00 CA 87 7A 5A 1C 81 CD CA 26 80 10 27
               43 8C 40 48 C9 97 58 1B 99 7A D9 45 D6 6A 70 CB

  0) Logout                              7) Ping host
  1) Assign interfaces                   8) Shell
  2) Set interface IP address            9) pfTop
  3) Reset the root password            10) Firewall log
  4) Reset to factory defaults          11) Reload all services
  5) Power off system                   12) Update from console
  6) Reboot system                      13) Restore a backup

Enter an option:
```

Tapez **1**

Tapez **N** à l'option "Do you want to configure LAGGS now ?"

Tapez **N** à l'option "Do you want to configure VLANs now ?"

Vous devrez choisir les interfaces pour le WAN et le LAN, c'est nos vSwitches nommés INTERNET et KIFORMATOU. Exécutez la commande suivante dans PowerShell Administrateur pour les identifier :

```
Get-VMNetworkAdapter -VMName "opnsense" |

    Select-Object VMName, Name, MacAddress
```

Vous devrez tomber sur l'affichage suivant (à adapter selon vos adresses MAC) :

```
VMName   Name            MacAddress

------   ----            ----------

opnsense Network Adapter 00155D00B72F

opnsense KIFORMATOU      00155D00B730
```

Tapez ensuite l'interface qui concerne chaque adresse MAC puis confirmez :

```
  5) Power off system                  12) Update from console
  6) Reboot system                     13) Restore a backup

Enter an option: 1

Do you want to configure LAGGs now? [y/N]: N
Do you want to configure VLANs now? [y/N]: N

Valid interfaces are:

hn0             00:15:5d:00:b7:2f Hyper-V Network Interface
hn1             00:15:5d:00:b7:30 Hyper-V Network Interface

If you do not know the names of your interfaces, you may choose to use
auto-detection. In that case, disconnect all interfaces now before
hitting 'a' to initiate auto detection.

Enter the WAN interface name or 'a' for auto-detection: hn0

Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(or nothing if finished): hn1

Enter the Optional interface 1 name or 'a' for auto-detection
(or nothing if finished):

The interfaces will be assigned as follows:

WAN  -> hn0
LAN  -> hn1

Do you want to proceed? [y/N]:
```

Nous allons ensuite modifier la configuration IP de chaque interface sachant qu'on veut du DHCP sur notre WAN et du statique sur le LAN. Tapez 2 puis configurez d'abord le LAN :

```
Available interfaces:

1 - LAN (hn1 - static, idassoc6)
2 - WAN (hn0 - dhcp, dhcp6)

Enter the number of the interface to configure: 1

Configure IPv4 address LAN interface via DHCP? [y/N] n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.0.1


Subnet masks are entered as bit counts (like CIDR notation).
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via WAN tracking? [Y/n] n
Configure IPv6 address LAN interface via DHCP6? [y/N] n

Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? [y/N] n
```

Configurez ensuite la vNIC “INTERNET” en DHCP :

```
Enter an option: 2

Available interfaces:

1 - LAN (hn1 - static)
2 - WAN (hn0 - dhcp, dhcp6)

Enter the number of the interface to configure: 2

Configure IPv4 address WAN interface via DHCP? [Y/n] y

Configure IPv6 address WAN interface via DHCP6? [Y/n] n

Enter the new WAN IPv6 address. Press <ENTER> for none:
>

Do you want to change the web GUI protocol from HTTPS to HTTP? [y/N] n
Do you want to generate a new self-signed web GUI certificate? [y/N] n
Restore web GUI access defaults? [y/N] n

Writing configuration...done.
Generating /etc/resolv.conf...done.
Generating /etc/hosts...done.
Configuring WAN interface...done.
Setting up routes for wan...done.
Starting Dnsmasq...done.
Starting Unbound DNS...done.
Configuring firewall.......done.
Starting Dnsmasq...done.
Starting Unbound DNS...done.
```

Nous pouvons ainsi accéder à l'interface web de OPNsense.

3.2. Accèder à l'interface Web

Avant d'accéder à l'interface Web de OPNsense, assurez-vous d'avoir la configuration IP suivante sur la vSwitch **"KIFORMATOU"** pour votre PC hôte :

```
Get-NetIPConfiguration -InterfaceAlias "vEthernet (KIFORMATOU)"


InterfaceAlias       : vEthernet (KIFORMATOU)
InterfaceIndex       : 13
InterfaceDescription : Hyper-V Virtual Ethernet Adapter #2
NetProfile.Name      : Réseau 5
IPv4Address          : 192.168.0.100
IPv6DefaultGateway   :
IPv4DefaultGateway   : 192.168.0.1
DNSServer            : 1.1.1.1
                       8.8.8.8
```

Si ce n'est pas le cas, exécutez les commandes suivantes avec **PowerShell Administrateur**:

```
$interfaceAlias = "Ethernet"

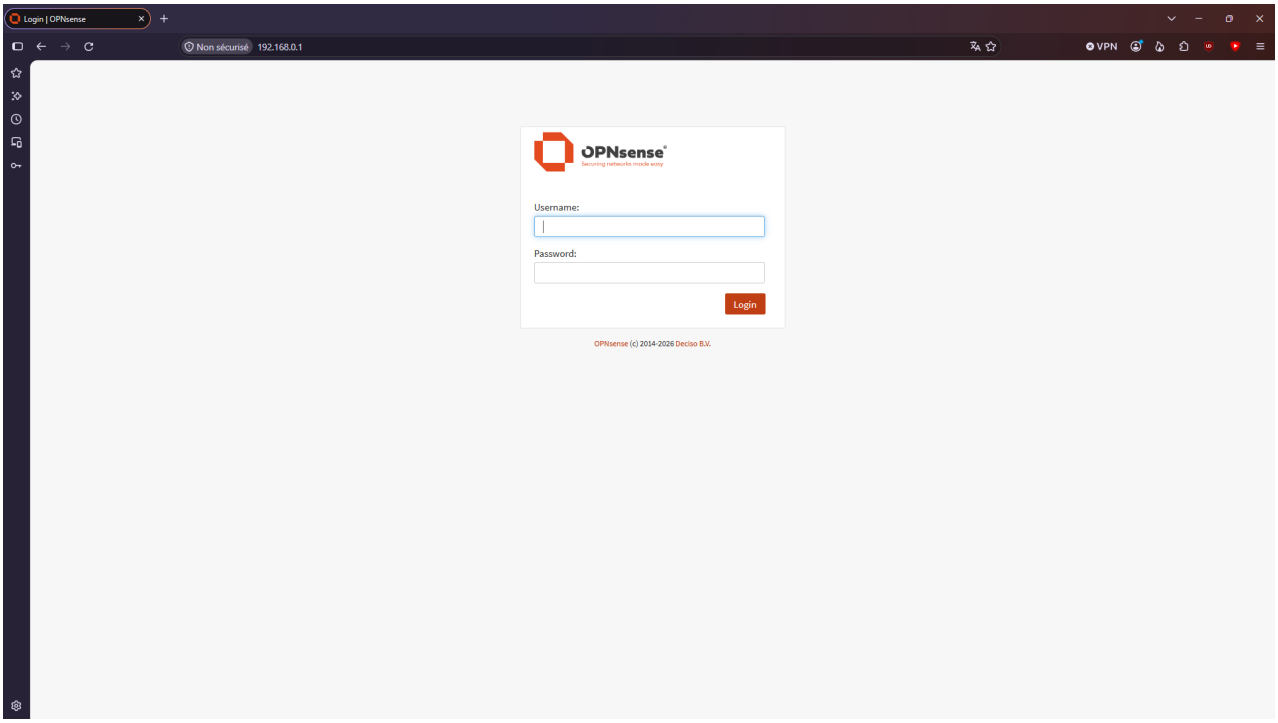

# Disable DHCP
Set-NetIPInterface -InterfaceAlias $interfaceAlias -AddressFamily IPv4 -
Dhcp Disabled


# Set the IPv4 address, subnet mask (/24 = 255.255.255.0), and default ga
teway
New-NetIPAddress -InterfaceAlias $interfaceAlias `
    -IPAddress "192.168.0.100" `
    -PrefixLength 24 `
    -DefaultGateway "192.168.0.1"

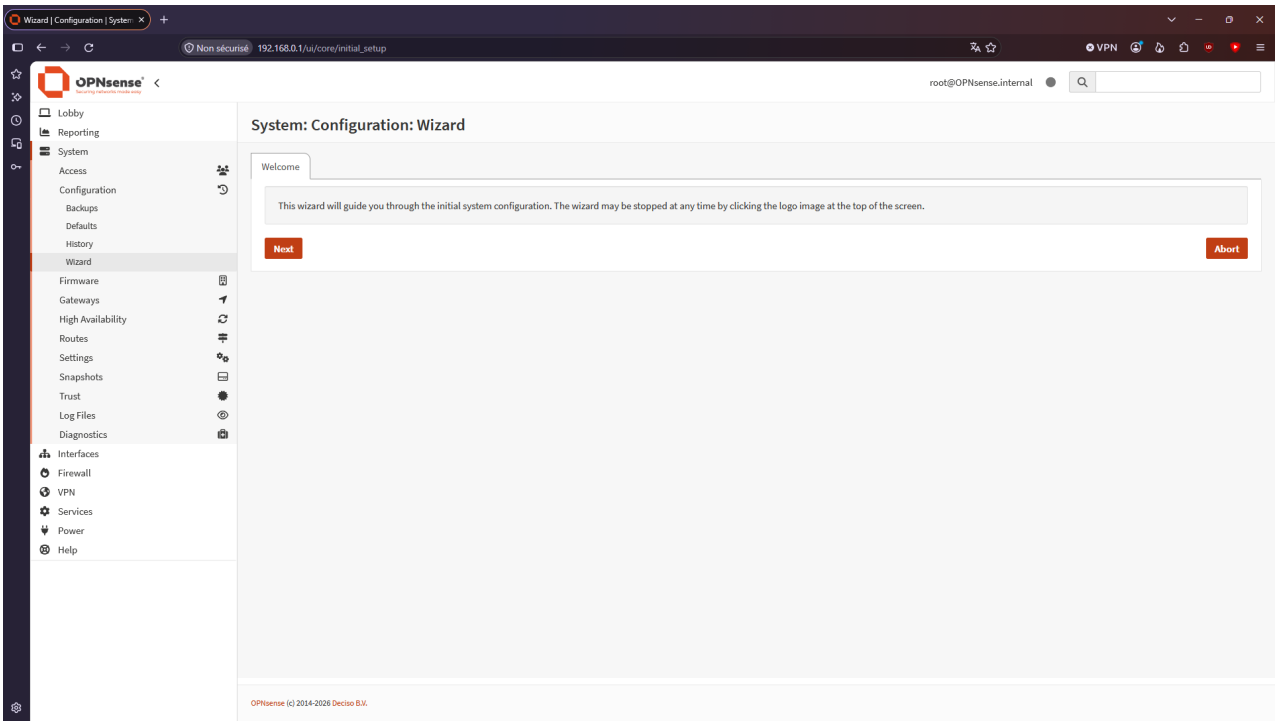

# Set DNS servers
Set-DnsClientServerAddress -InterfaceAlias $interfaceAlias `
    -ServerAddresses ("1.1.1.1", "8.8.8.8")

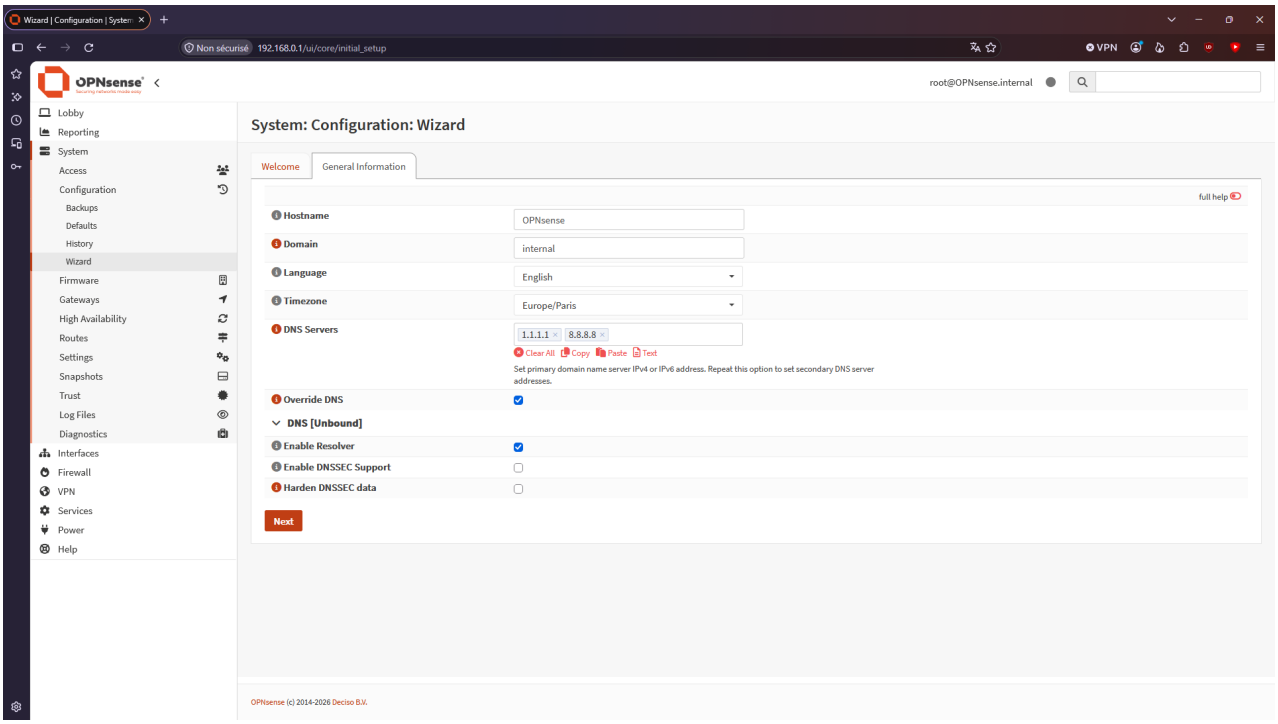

# Verify the network configuration
Get-NetIPConfiguration -InterfaceAlias $interfaceAlias
```

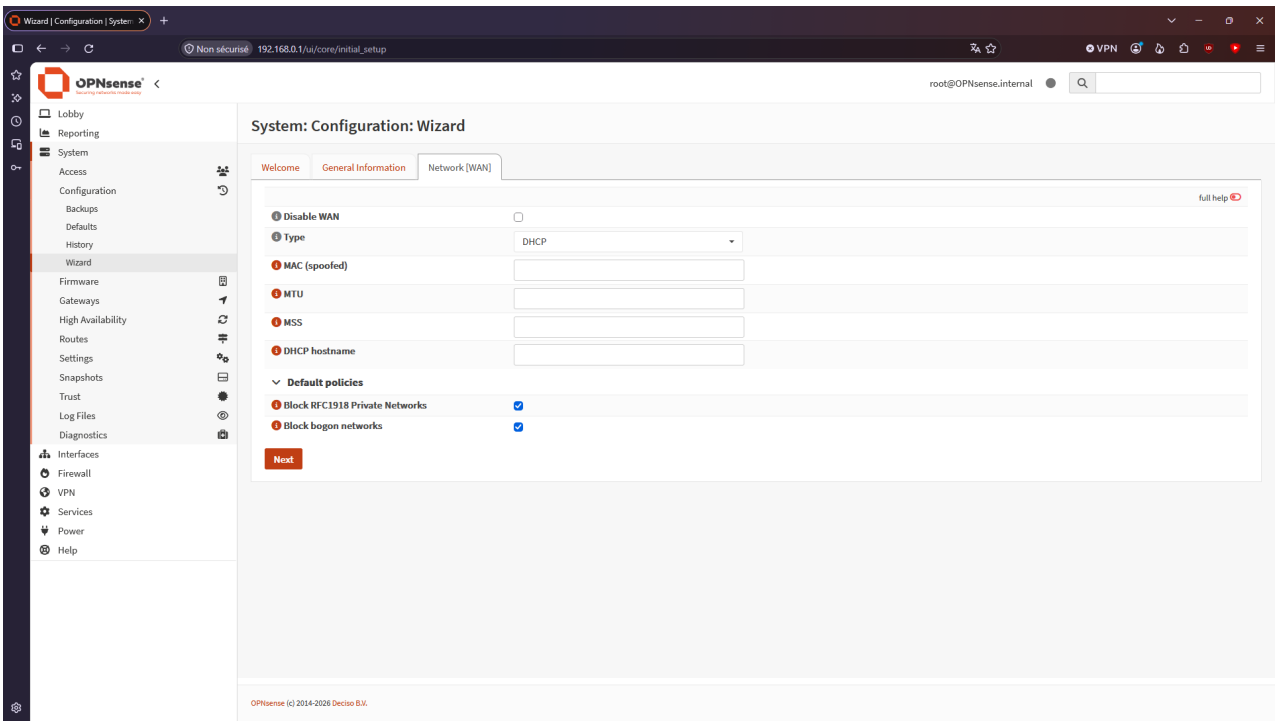
Connectez-vous ensuite avec le navigateur web de votre choix à l'adresse https://192.168.0.1. Vous devrez tomber sur la fenêtre suivante :

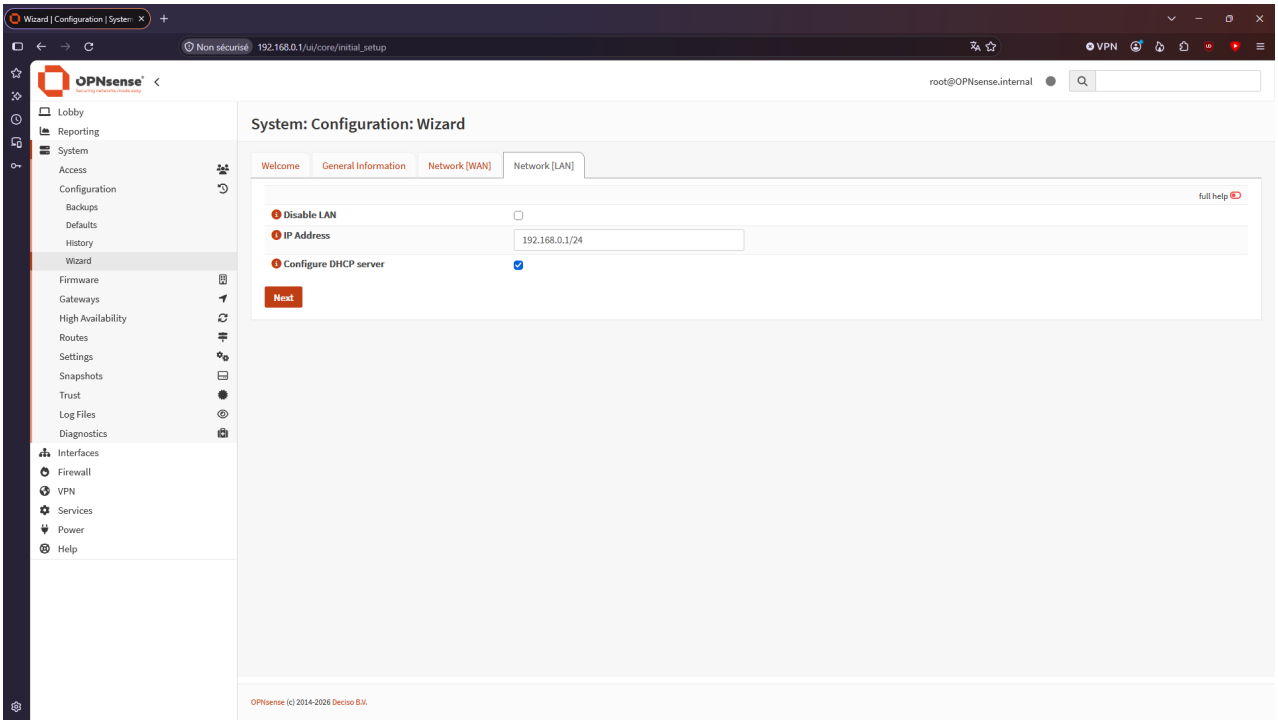


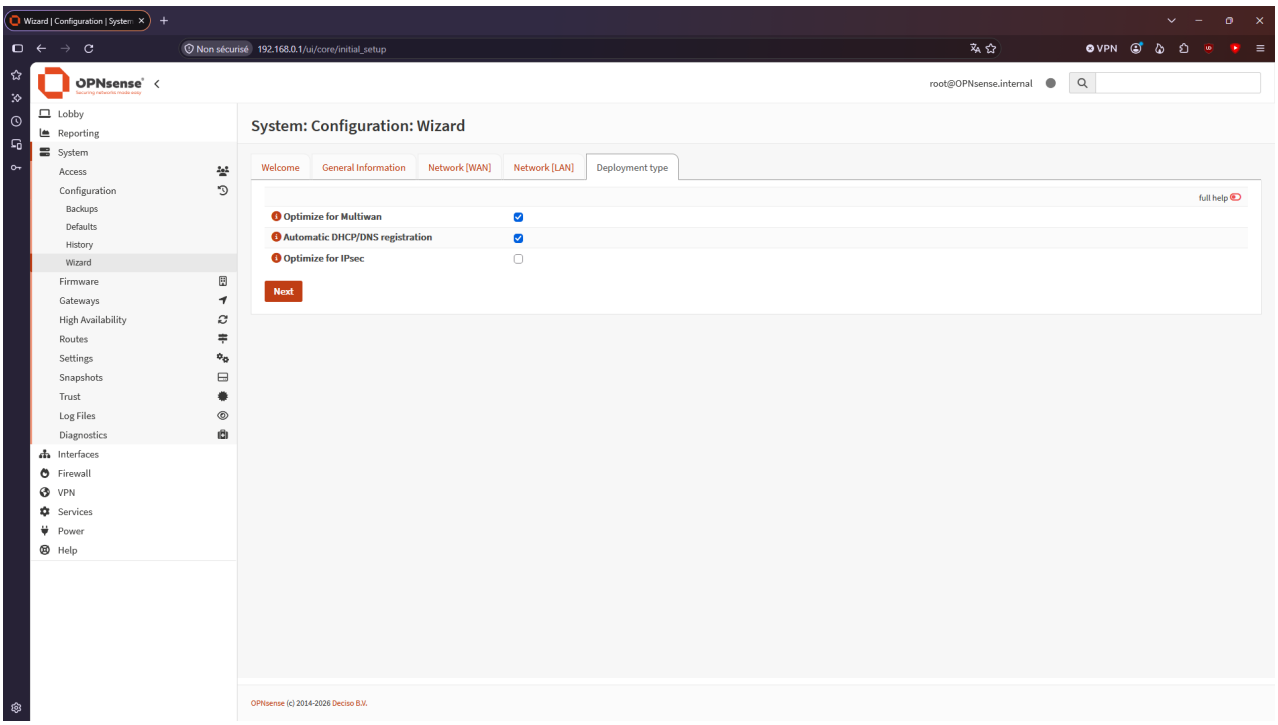
Rentrez vos identifiants pour le root puis suivez la configuration initale à l'arrivée :

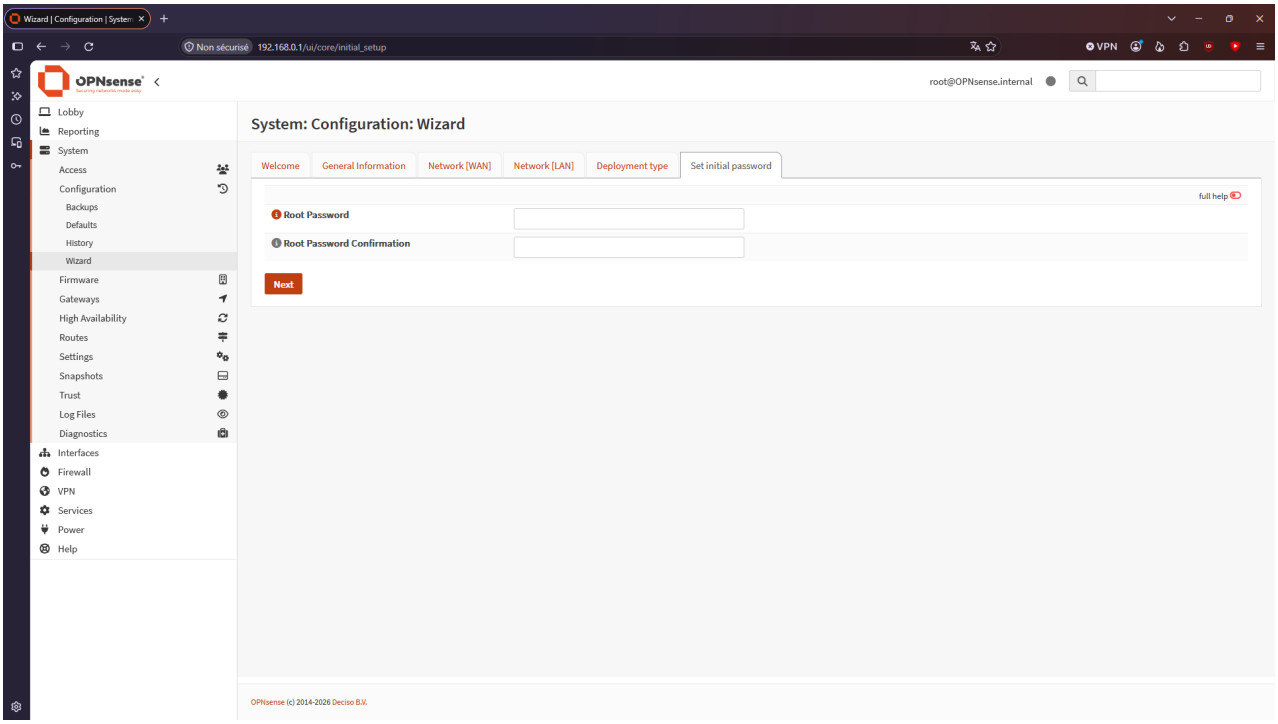


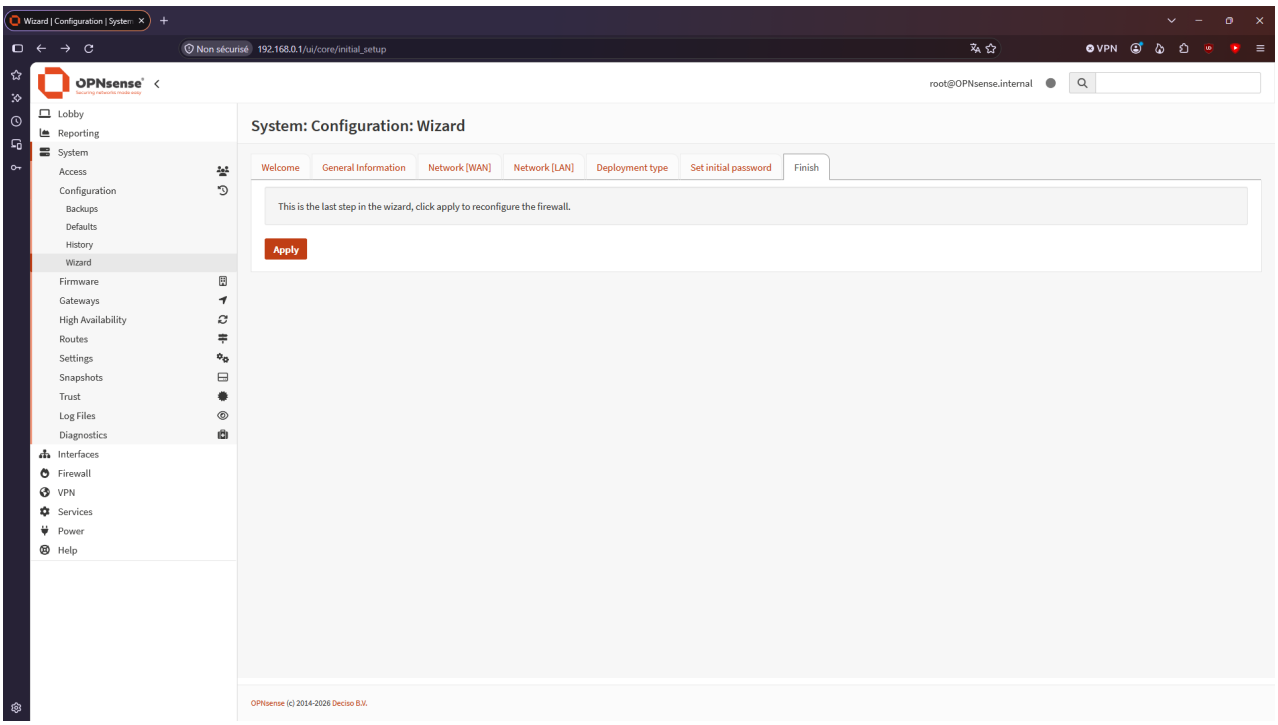


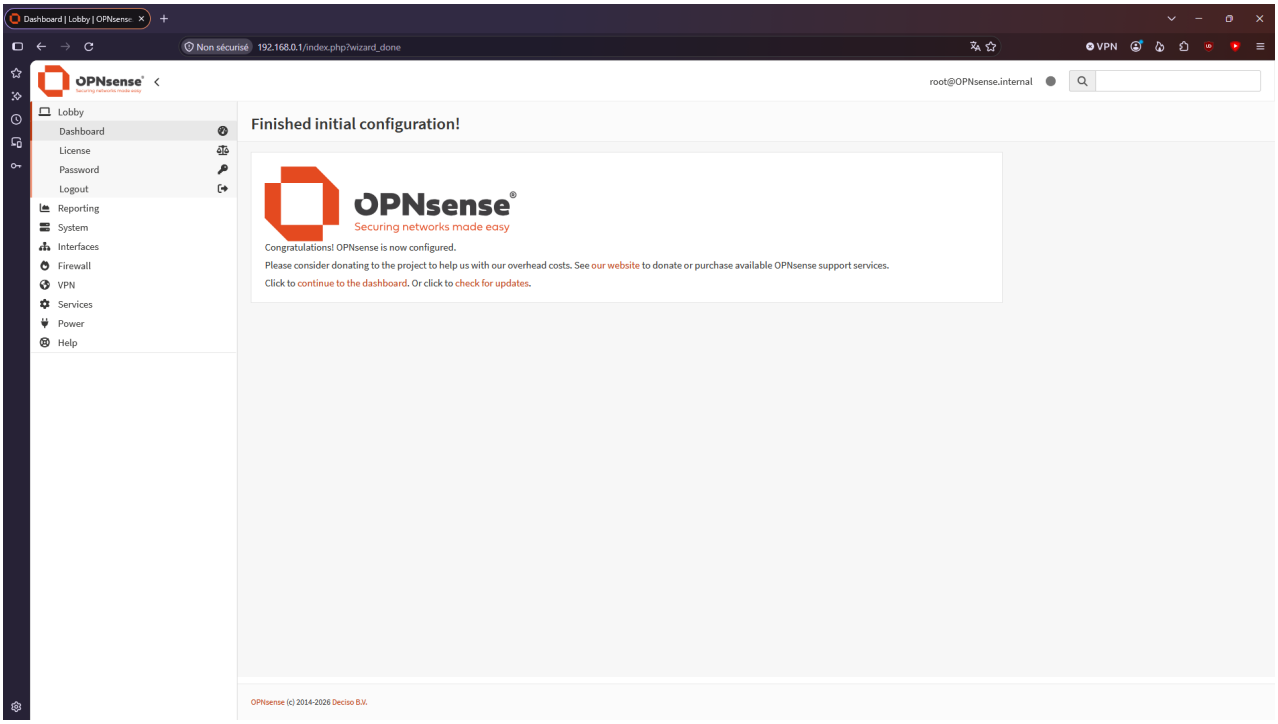

Wizard | Configuration | System
Non sécurisé 192.168.0.1/ui/core/initial_setup
VPN
OPNsense
root@OPNsense.internal
Lobby
Reporting
System
Access
Configuration
Backups
Defaults
History
Wizard
Firmware
Gateways
High Availability
Routes
Settings
Snapshots
Trust
Log Files
Diagnostics
Interfaces
Firewall
VPN
Services
Power
Help
System: Configuration: Wizard
Welcome
General Information
Network [WAN]
full help
Disable WAN
Type
DHCP
MAC (spoofed)
MTU
MSS
DHCP hostname
Default policies
Block RFC1918 Private Networks
Block bogon networks
Next
OPNsense (c) 2014-2026 Deciso B.V.


Wizard | Configuration | System
Non sécurisé 192.168.0.1/ui/core/initial_setup
VPN
OPNsense
root@OPNsense.internal
Lobby
Reporting
System
Access
Configuration
Backups
Defaults
History
Wizard
Firmware
Gateways
High Availability
Routes
Settings
Snapshots
Trust
Log Files
Diagnostics
Interfaces
Firewall
VPN
Services
Power
Help
System: Configuration: Wizard
Welcome
General Information
Network [WAN]
Network [LAN]
full help
Disable LAN
IP Address
192.168.0.1/24
Configure DHCP server
Next
OPNsense (c) 2014-2026 Deciso B.V.

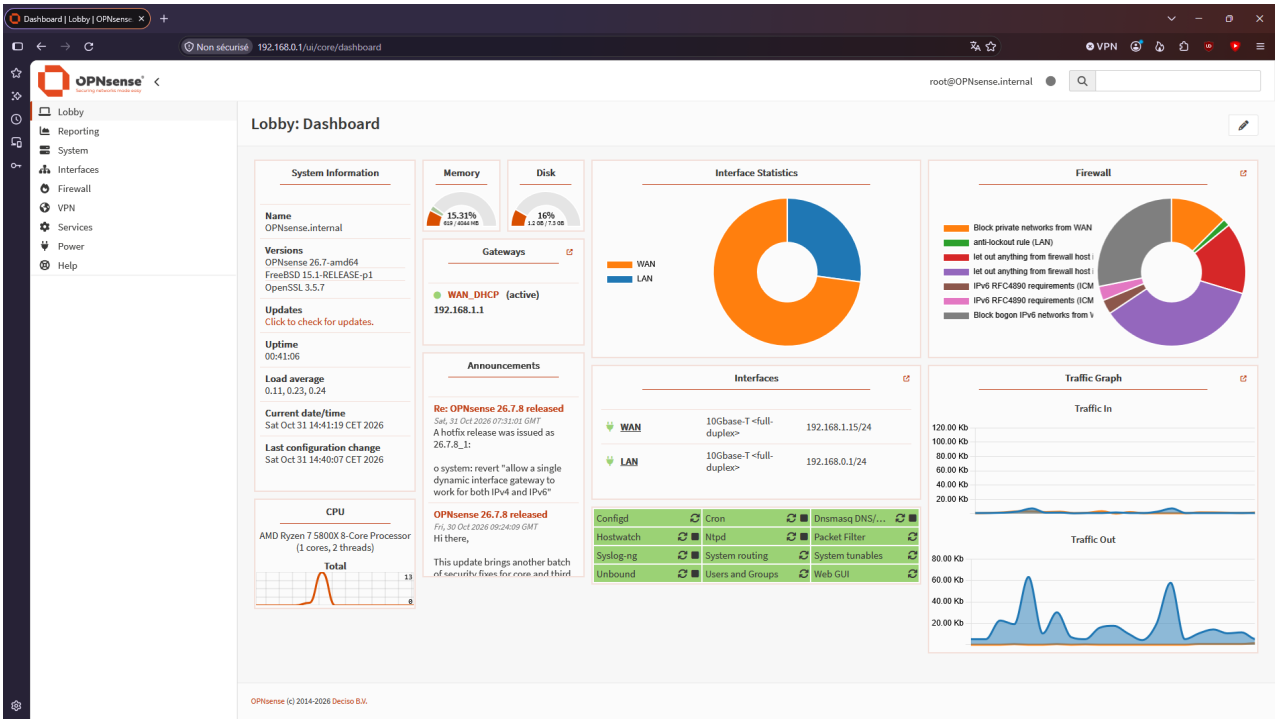

Wizard | Configuration | System
Non sécurisé 192.168.0.1/ui/core/initial_setup
VPN
OPNsense
root@OPNsense.internal
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Wizard
Firmware
Gateways
High Availability
Routes
Settings
Snapshots
Trust
Log Files
Diagnostics
Interfaces
Firewall
VPN
Services
Power
Help
System: Configuration: Wizard
Welcome
General Information
Network [WAN]
Network [LAN]
Deployment type
full help
Optimize for Multiwan
Automatic DHCP/DNS registration
Optimize for IPsec
Next
OPNsense (c) 2014-2026 Deciso B.V.


Wizard | Configuration | System
Non sécurisé 192.168.0.1/ui/core/initial_setup
VPN
OPNsense
root@OPNsense.internal
Lobby
Reporting
System
Access
Configuration
Backups
Defaults
History
Wizard
Firmware
Gateways
High Availability
Routes
Settings
Snapshots
Trust
Log Files
Diagnostics
Interfaces
Firewall
VPN
Services
Power
Help
System: Configuration: Wizard
Welcome
General Information
Network [WAN]
Network [LAN]
Deployment type
Set initial password
full help
Root Password
Root Password Confirmation
Next
OPNsense (c) 2014-2026 Deciso B.V.

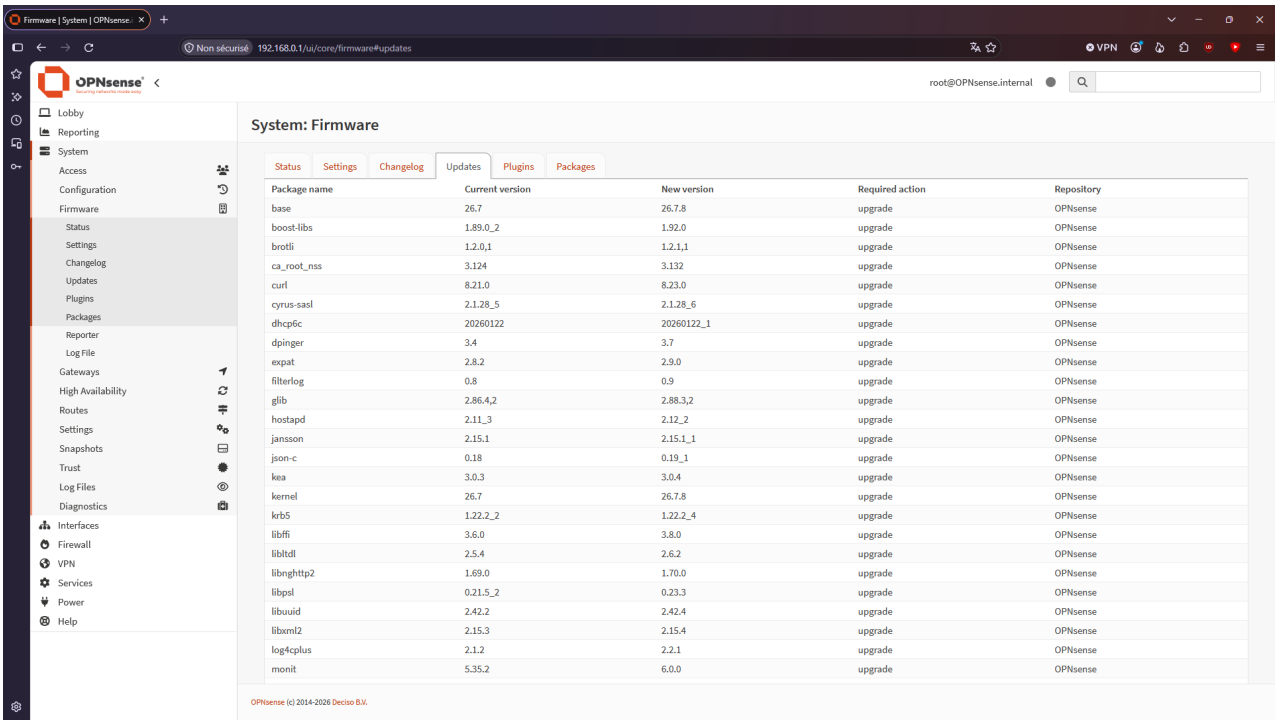




Terminez la configuration initale et revenir dans **Lobby/Dashboard** pour confirmez les informations.



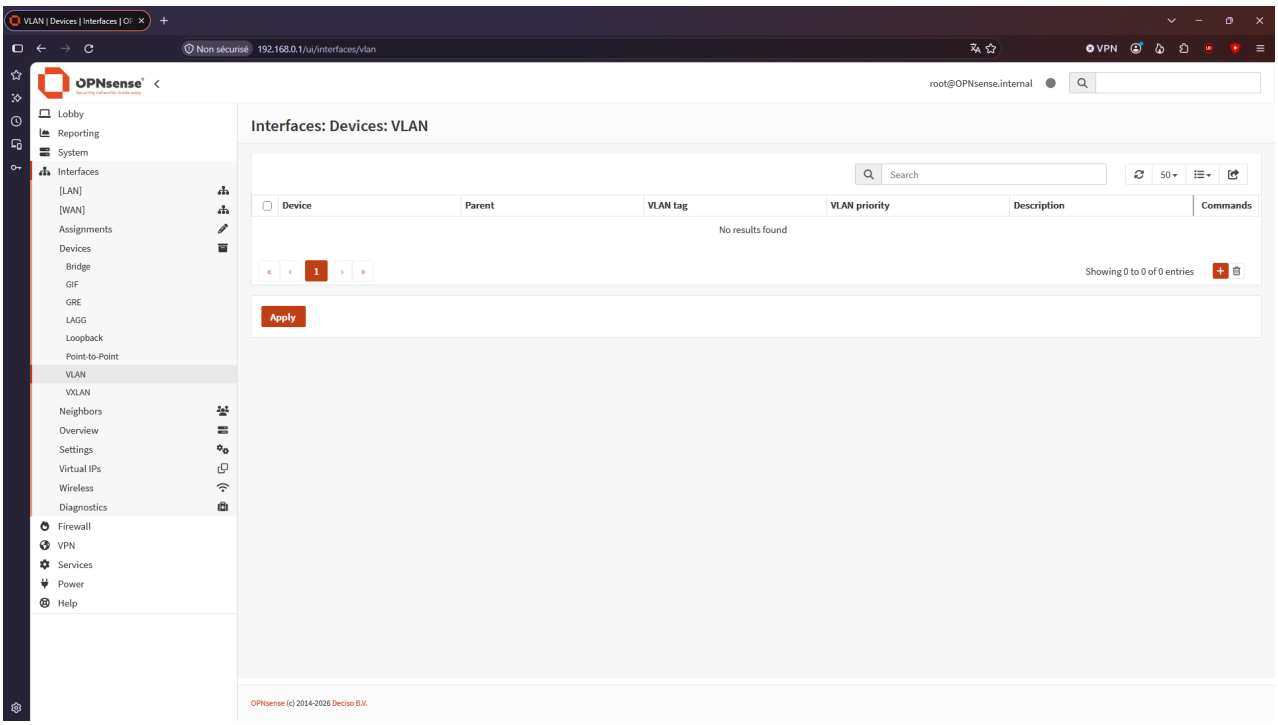
Mettez à jour OPNsense en allant dans System/Firmware/Updates



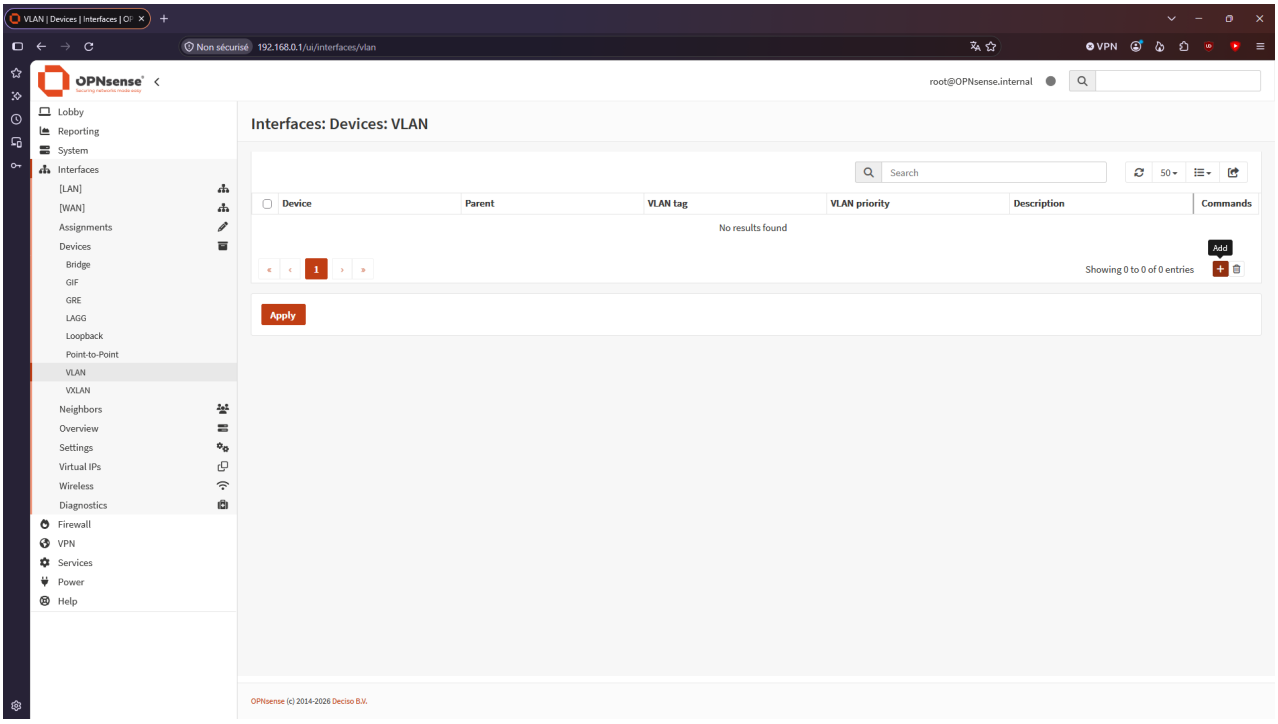
Appliquez les MAJ. Vous avez donc terminé l'installation et la configuration de OPNsense. Vous pouvez également créer les VLANs (optionnel) et des serveurs DHCP. N'oubliez pas d'exporter la VM pour la conserver comme template du projet.

3.3. Créer les VLAN

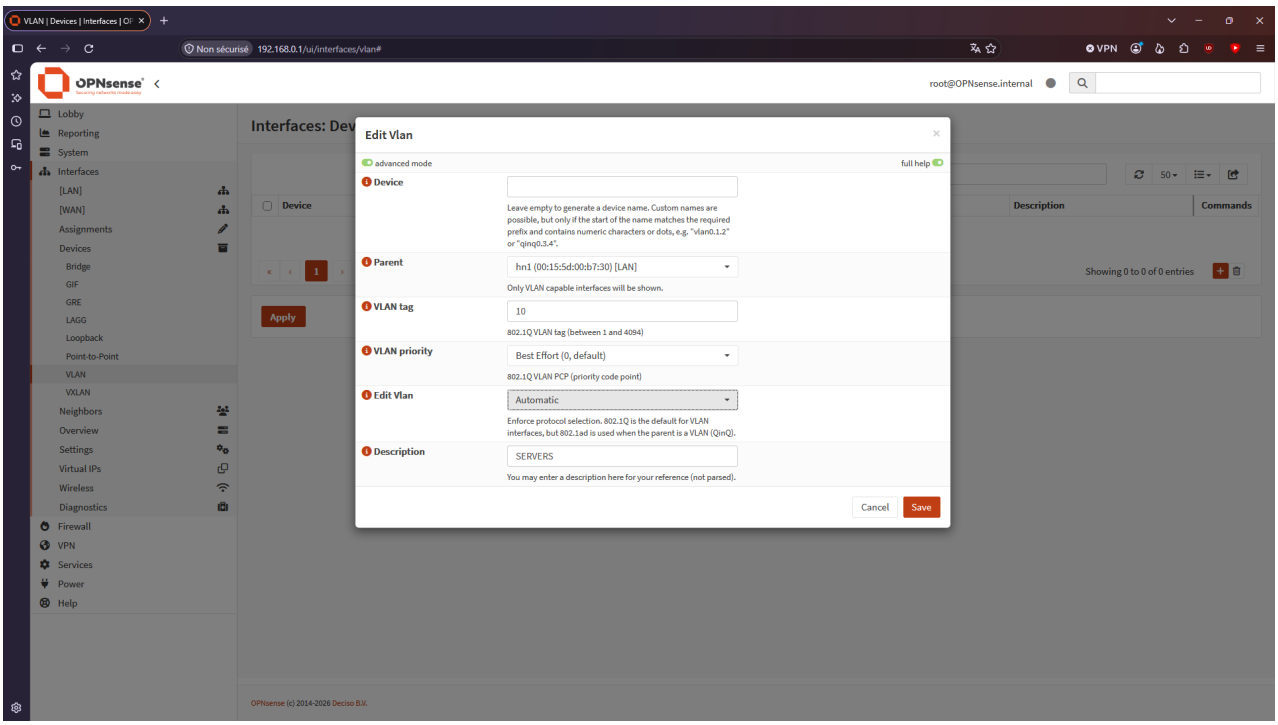
Nous allons créer les VLAN 10 (SERVERS), 20 (TEACHERS) et 30 (STUDENTS) pour notre infrastructure. Allez dans Interfaces/VLAN sur l'interface web de OPNsense :

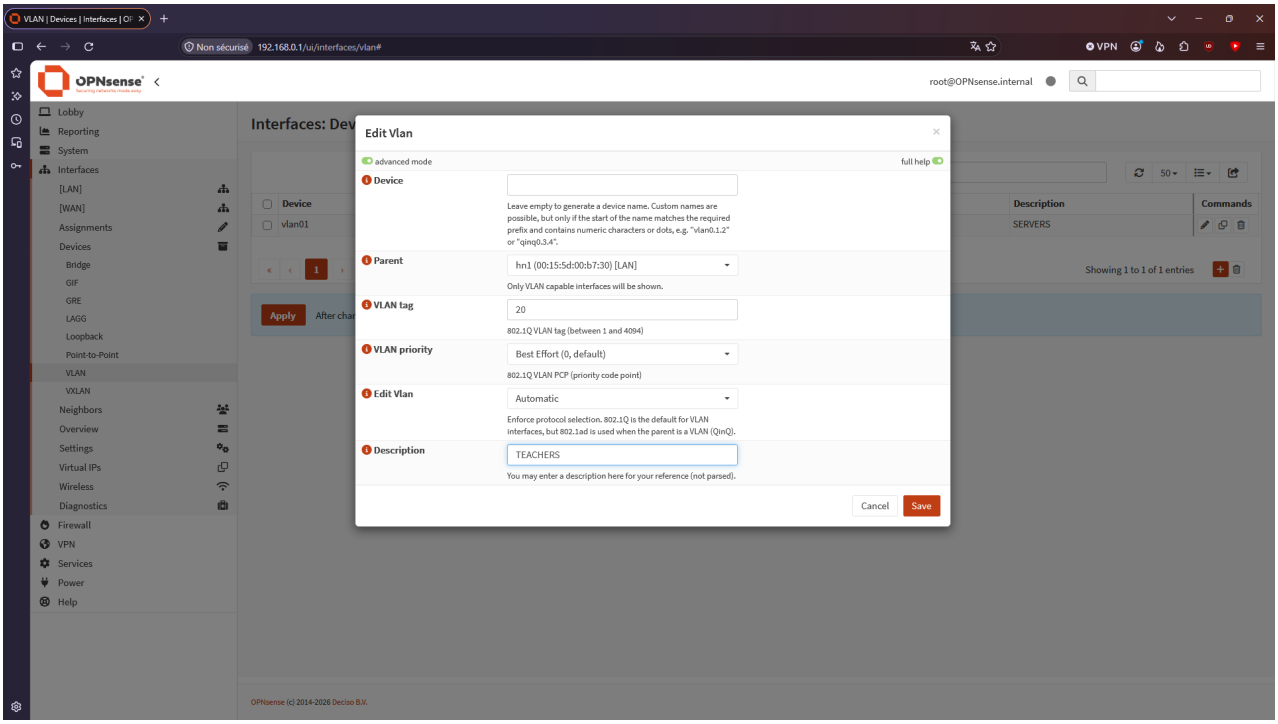


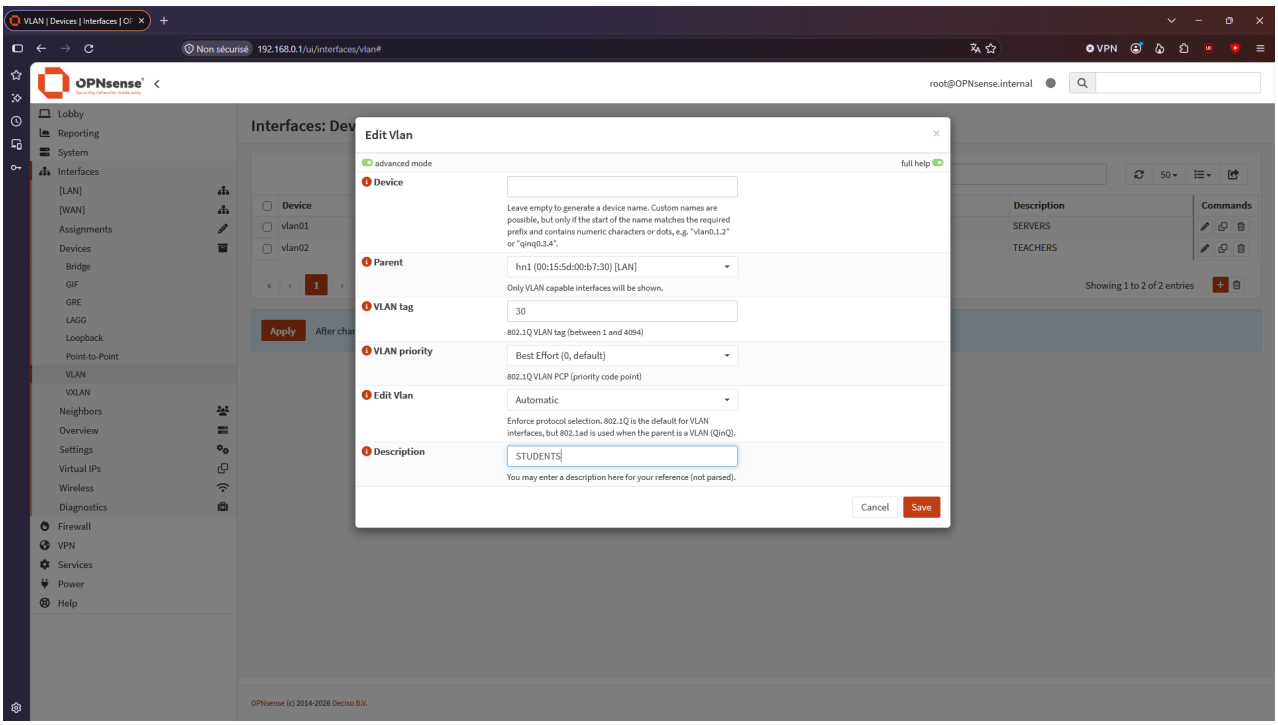
Cliquez sur le bouton **+ (Add)** pour ajouter un VLAN :

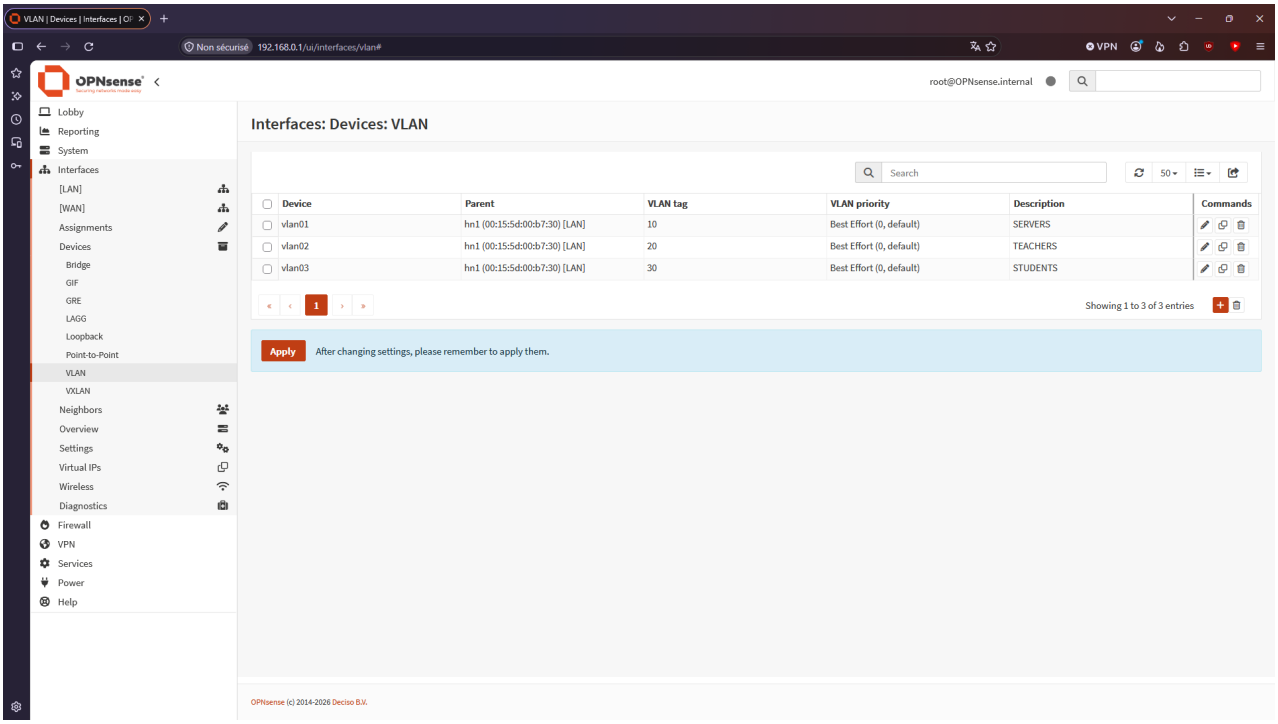


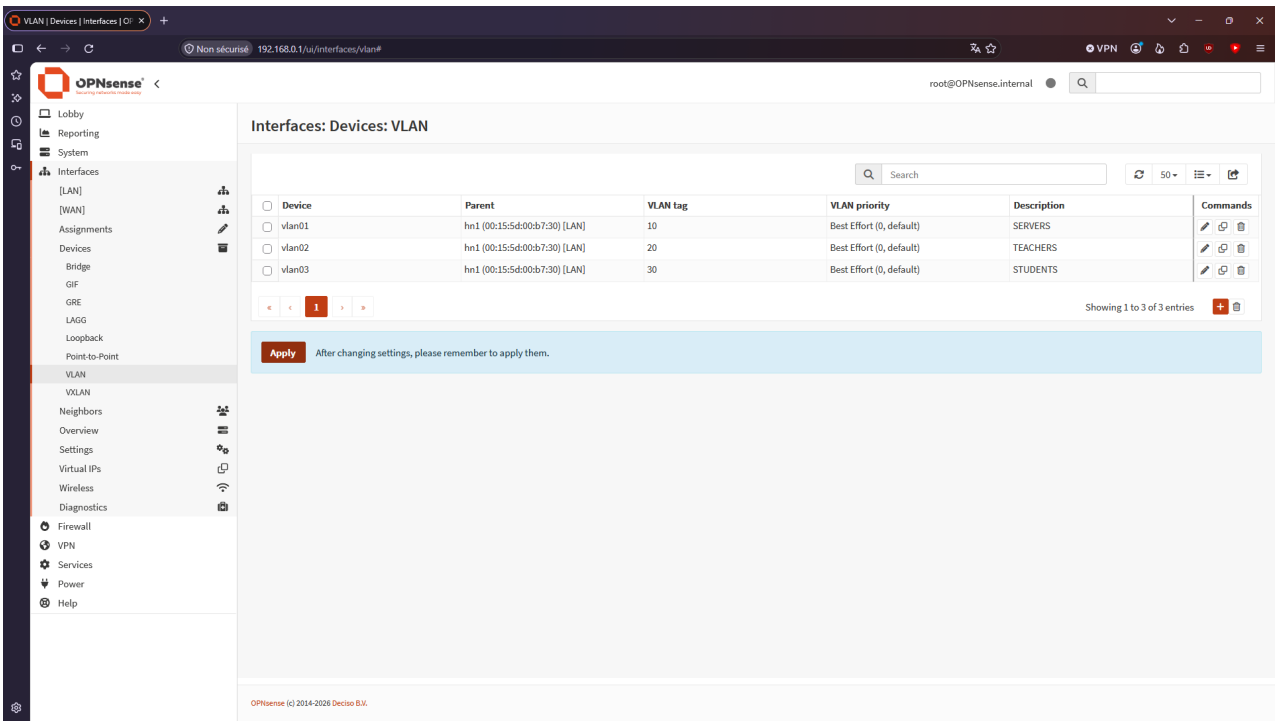
Ajoutez les paramètres suivants pour chaque création de VLAN puis cliquez sur Save :


Edit Vlan
advanced mode
full help
Device
Leave empty to generate a device name. Custom names are possible, but only if the start of the name matches the required prefix and contains numeric characters or dots, e.g. "vlan0.1.2" or "qinq0.3.4".
Parent
hn1 (00:15:5d:00:b7:30) [LAN]
Only VLAN capable interfaces will be shown.
VLAN tag
10
802.1Q VLAN tag (between 1 and 4094)
VLAN priority
Best Effort (0, default)
802.1Q VLAN PCP (priority code point)
Edit Vlan
Automatic
Enforce protocol selection. 802.1Q is the default for VLAN interfaces, but 802.1ad is used when the parent is a VLAN (QinQ).
Description
SERVERS
You may enter a description here for your reference (not parsed).
Cancel
Save

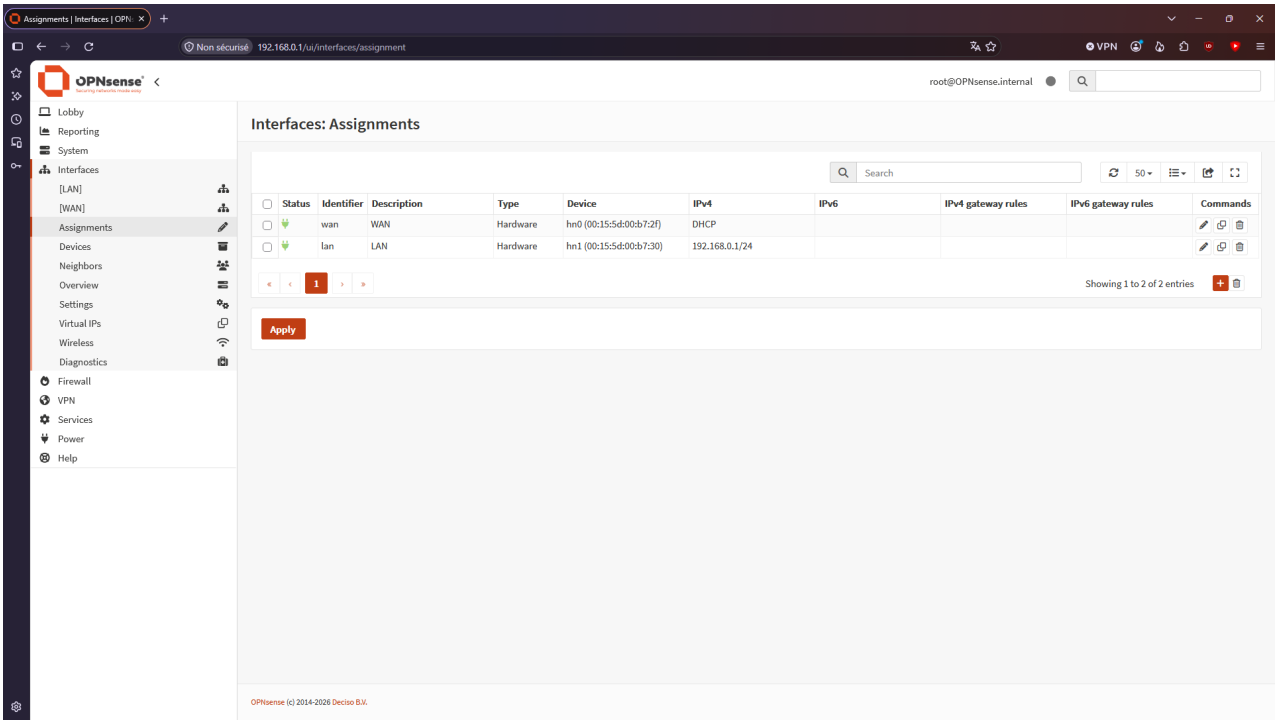

Edit Vlan
advanced mode
full help
Device
Leave empty to generate a device name. Custom names are possible, but only if the start of the name matches the required prefix and contains numeric characters or dots, e.g. "vlan0.1.2" or "qinq0.3.4".
Parent
hn1 (00:15:5d:00:b7:30) [LAN]
Only VLAN capable interfaces will be shown.
VLAN tag
20
802.1Q VLAN tag (between 1 and 4094)
VLAN priority
Best Effort (0, default)
802.1Q VLAN PCP (priority code point)
Edit Vlan
Automatic
Enforce protocol selection. 802.1Q is the default for VLAN interfaces, but 802.1ad is used when the parent is a VLAN (QinQ).
Description
TEACHERS
You may enter a description here for your reference (not parsed).
Cancel
Save


Edit Vlan
advanced mode
full help
Device
Leave empty to generate a device name. Custom names are possible, but only if the start of the name matches the required prefix and contains numeric characters or dots, e.g. "vlan0.1.2" or "qinq0.3.4".
Parent
hn1 (00:15:5d:00:b7:30) [LAN]
Only VLAN capable interfaces will be shown.
VLAN tag
30
802.1Q VLAN tag (between 1 and 4094)
VLAN priority
Best Effort (0, default)
802.1Q VLAN PCP (priority code point)
Edit Vlan
Automatic
Enforce protocol selection. 802.1Q is the default for VLAN interfaces, but 802.1ad is used when the parent is a VLAN (QinQ).
Description
STUDENTS
You may enter a description here for your reference (not parsed).
Cancel
Save

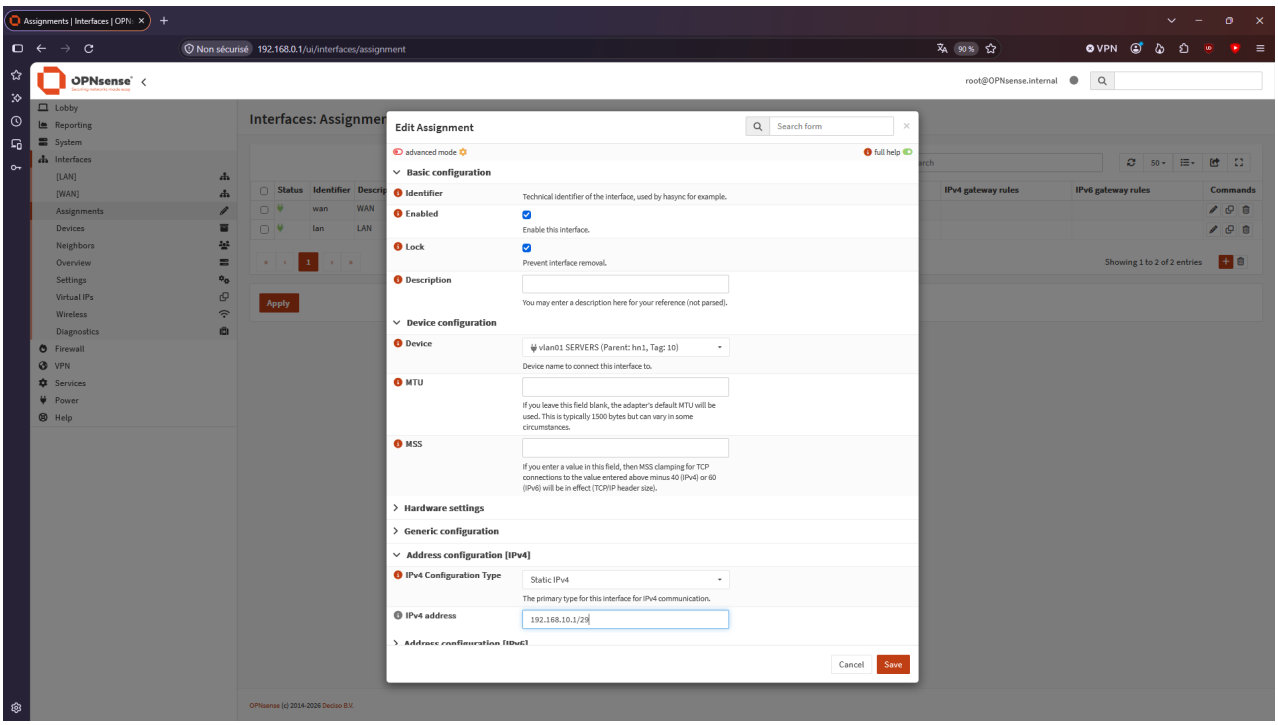

Interfaces: Devices: VLAN
Search
Device
Parent
VLAN tag
VLAN priority
Description
Commands
vlan01
hn1 (00:15:5d:00:b7:30) [LAN]
10
Best Effort (0, default)
SERVERS
vlan02
hn1 (00:15:5d:00:b7:30) [LAN]
20
Best Effort (0, default)
TEACHERS
vlan03
hn1 (00:15:5d:00:b7:30) [LAN]
30
Best Effort (0, default)
STUDENTS
Showing 1 to 3 of 3 entries
Apply
After changing settings, please remember to apply them.

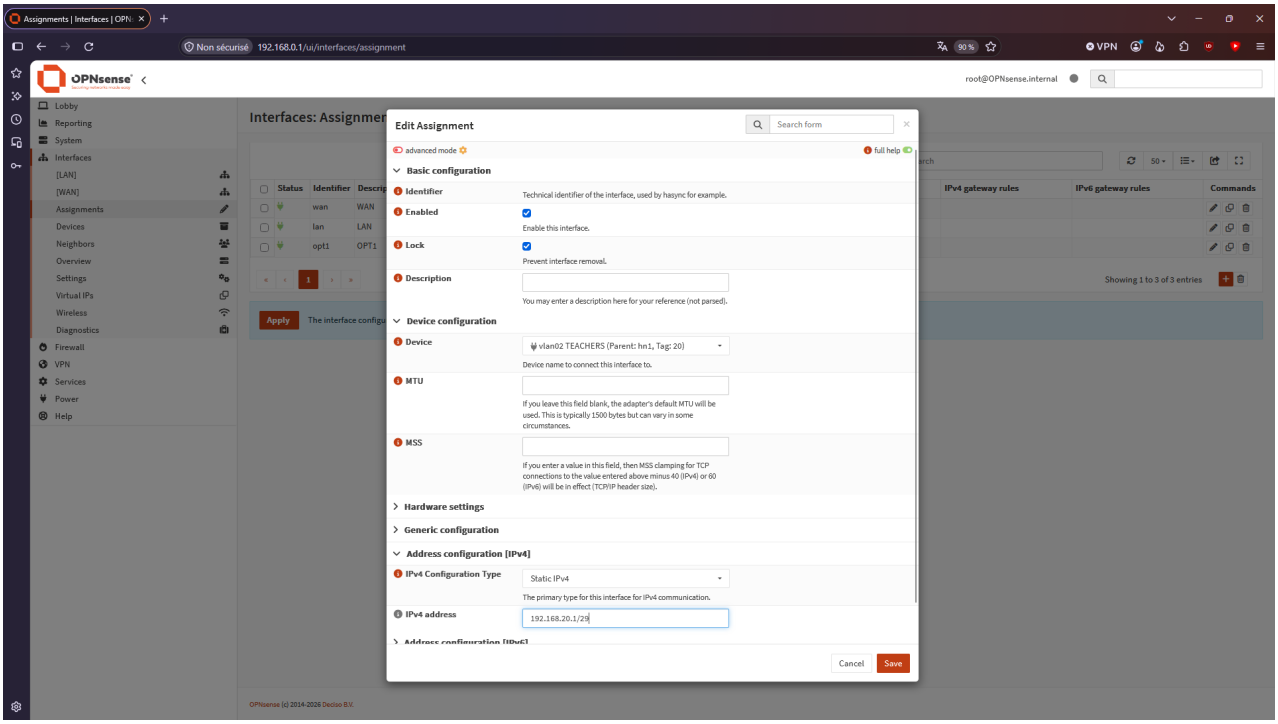


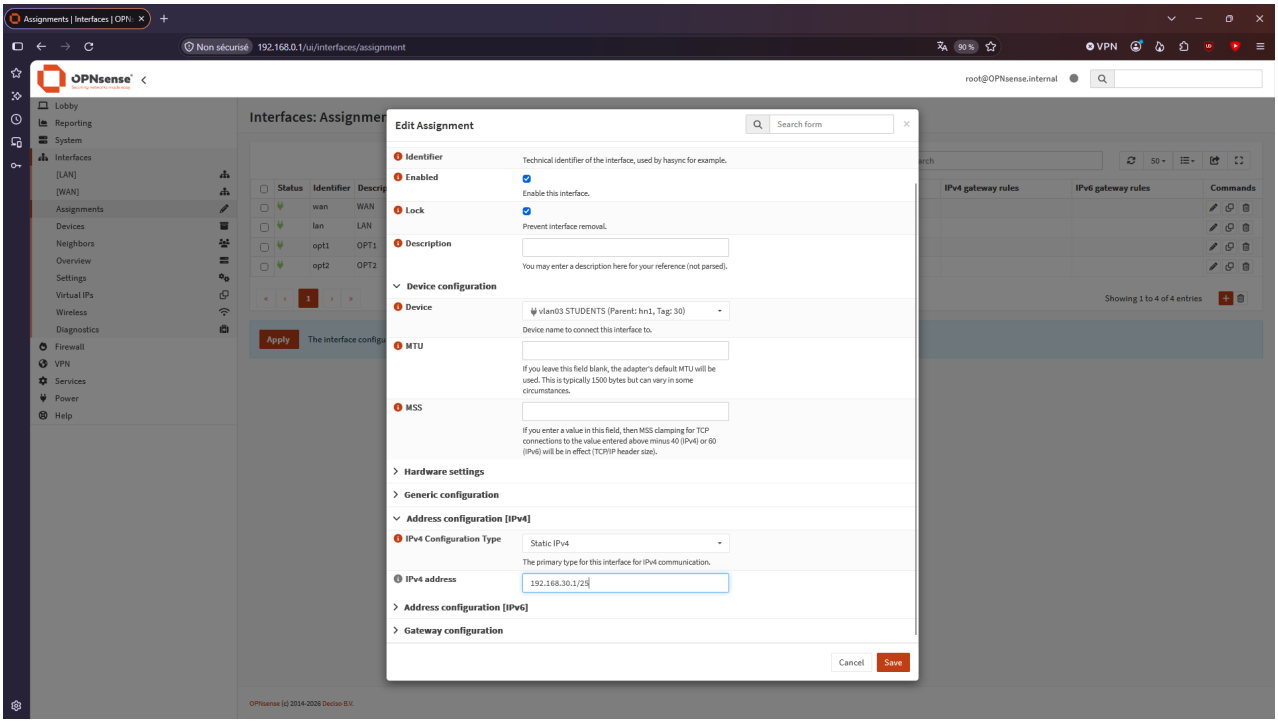
Appliquez les modifications en cliquant sur **Apply**. Allez ensuite dans Interfaces/Assignments :

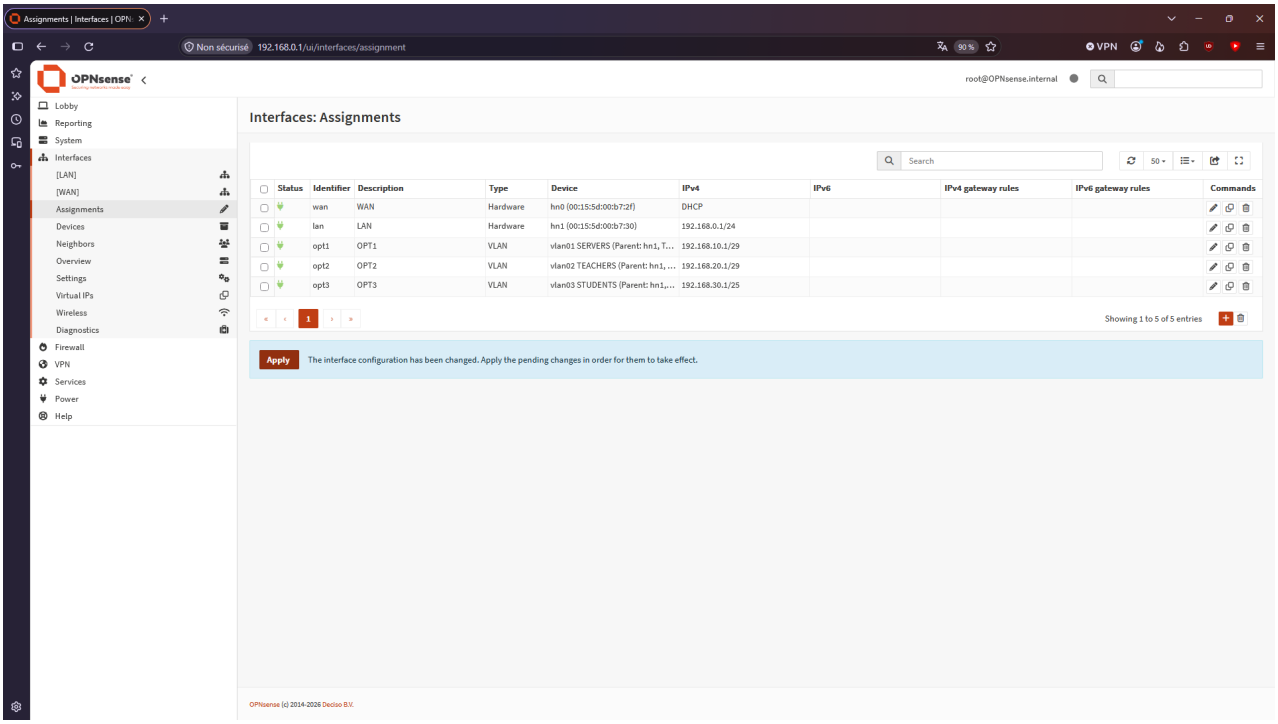


Affectez et activer les interfaces pour chaque VLAN avec une adresse statique :

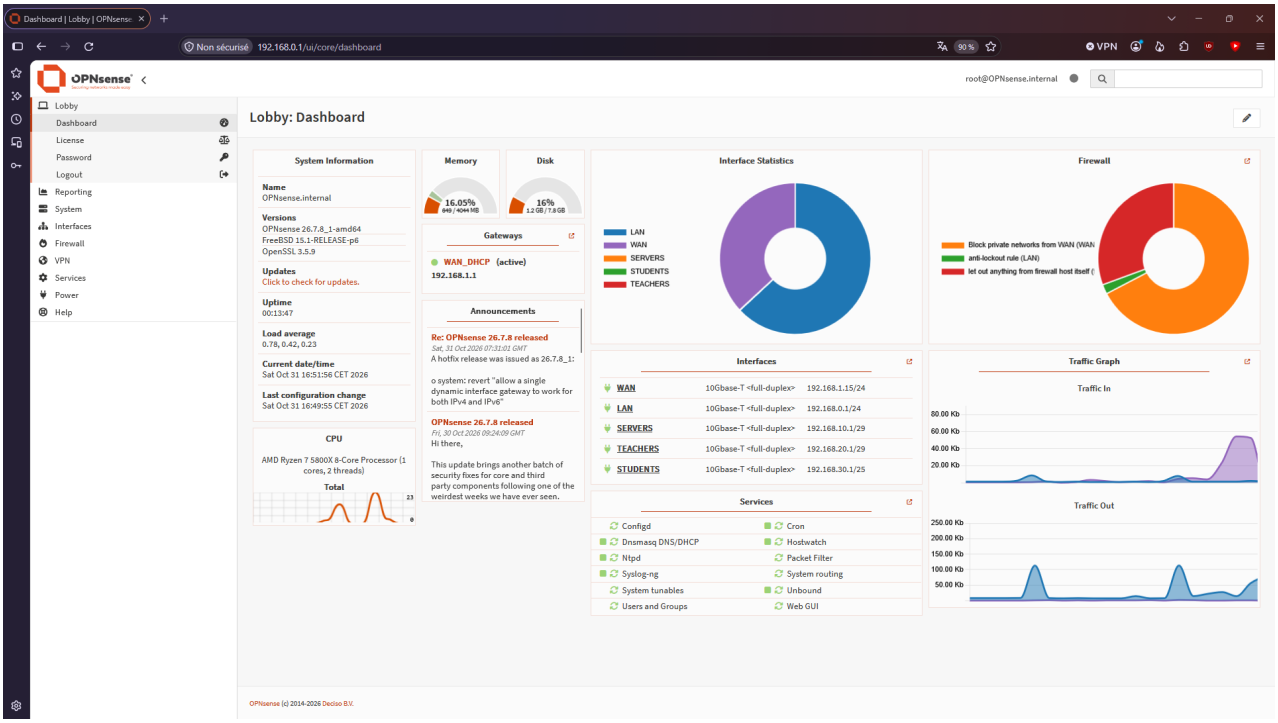

Edit Assignment
Search form
advanced mode
full help
Basic configuration
Identifier
Technical identifier of the interface, used by hasync for example.
Enabled
Enable this interface.
Lock
Prevent interface removal.
Description
You may enter a description here for your reference (not parsed).
Device configuration
Device
vlan01 SERVERS (Parent: hn1, Tag: 10)
Device name to connect this interface to.
MTU
If you leave this field blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.
MSS
If you enter a value in this field, then MSS clamping for TCP connections to the value entered above minus 40 (IPv4) or 60 (IPv6) will be in effect (TCP/IP header size).
Hardware settings
Generic configuration
Address configuration [IPv4]
IPv4 Configuration Type
Static IPv4
The primary type for this interface for IPv4 communication.
IPv4 address
192.168.10.1/29
Cancel
Save


Edit Assignment
Search form
advanced mode
full help
Basic configuration
Identifier
Technical identifier of the interface, used by hasync for example.
Enabled
Enable this interface.
Lock
Prevent interface removal.
Description
You may enter a description here for your reference (not parsed).
Device configuration
Device
vlan02 TEACHERS (Parent: hn1, Tag: 20)
Device name to connect this interface to.
MTU
If you leave this field blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.
MSS
If you enter a value in this field, then MSS clamping for TCP connections to the value entered above minus 40 (IPv4) or 60 (IPv6) will be in effect (TCP/IP header size).
Hardware settings
Generic configuration
Address configuration [IPv4]
IPv4 Configuration Type
Static IPv4
The primary type for this interface for IPv4 communication.
IPv4 address
192.168.20.1/29
Cancel
Save





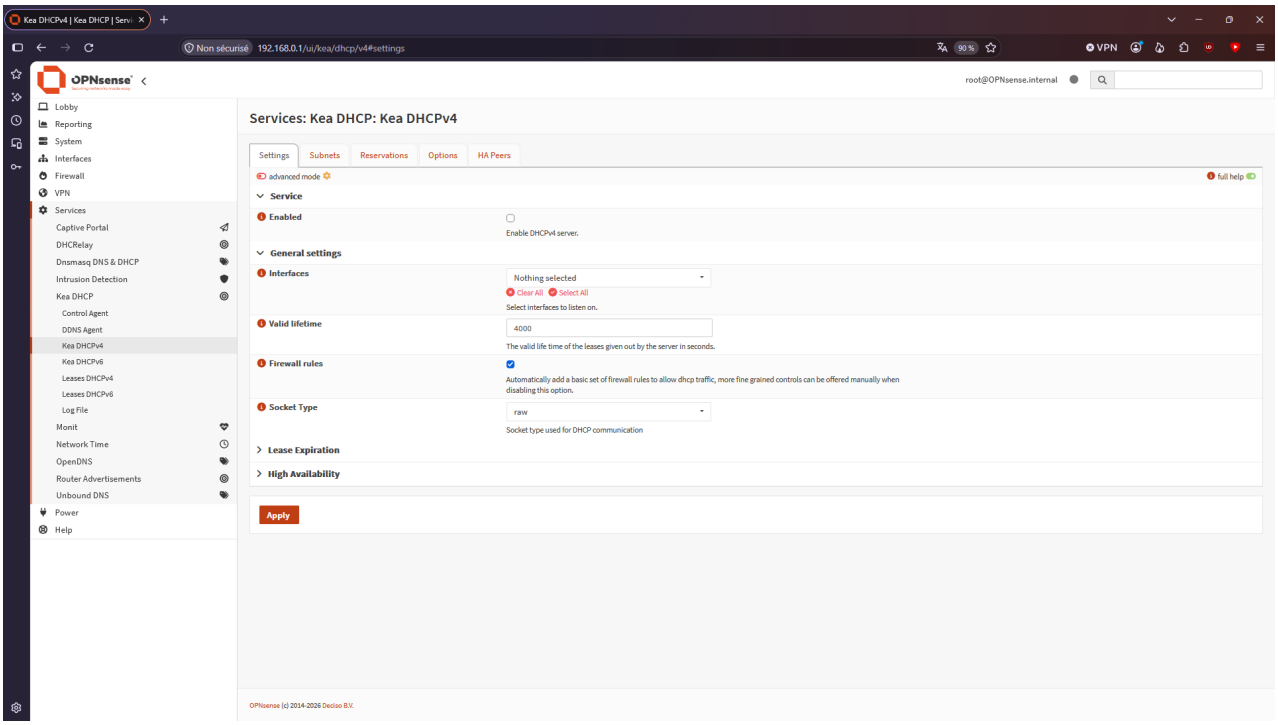
Appliquez ensuite les modifications (vous pouvez également ajoutez une description pour chaque VLAN comme SERVERS). En retournant dans **Lobby/Dashbord**, vous devez retrouver les interfaces de chaque VLAN :



Nous avons terminé avec la création des VLAN.

3.4. Créer les serveurs DHCP

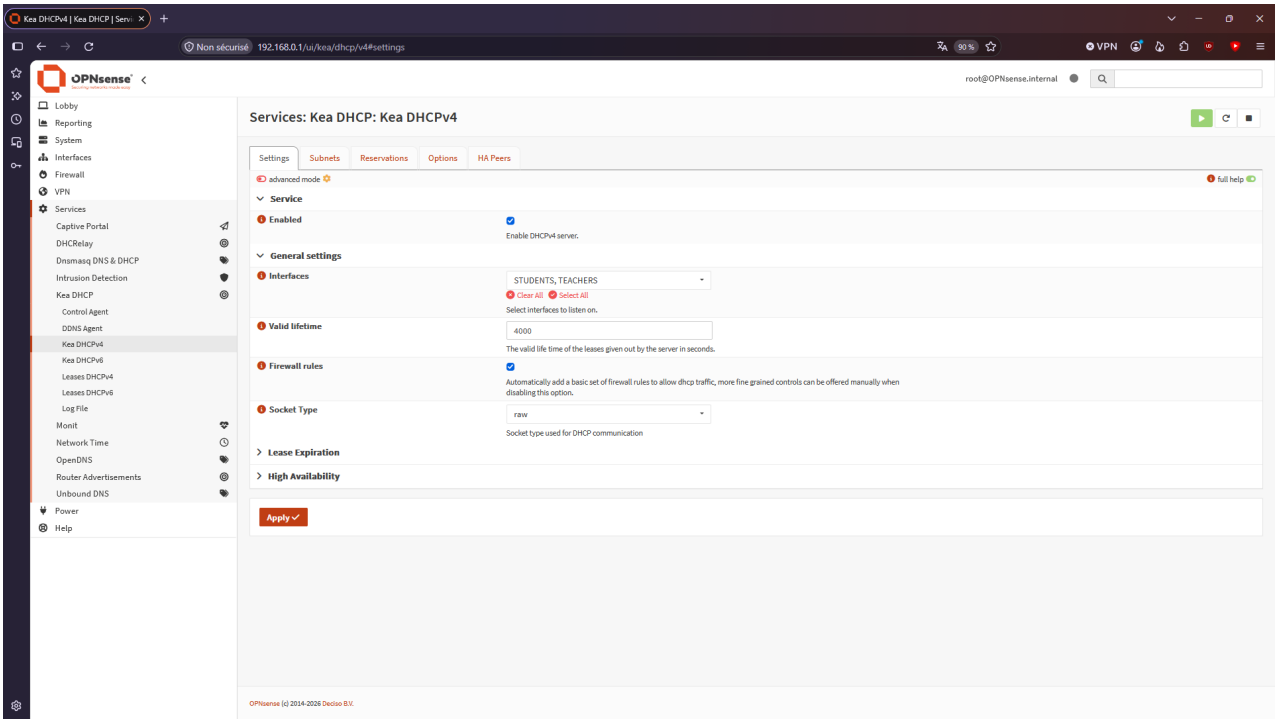
Nous allons après créer les serveurs DHCP pour les VLAN 20 et 30. Le VLAN 10 contient uniquement des serveurs avec une adresse IPv4 statique. Allez dans Services/Kea DHCP/Kea DHCPv4 :



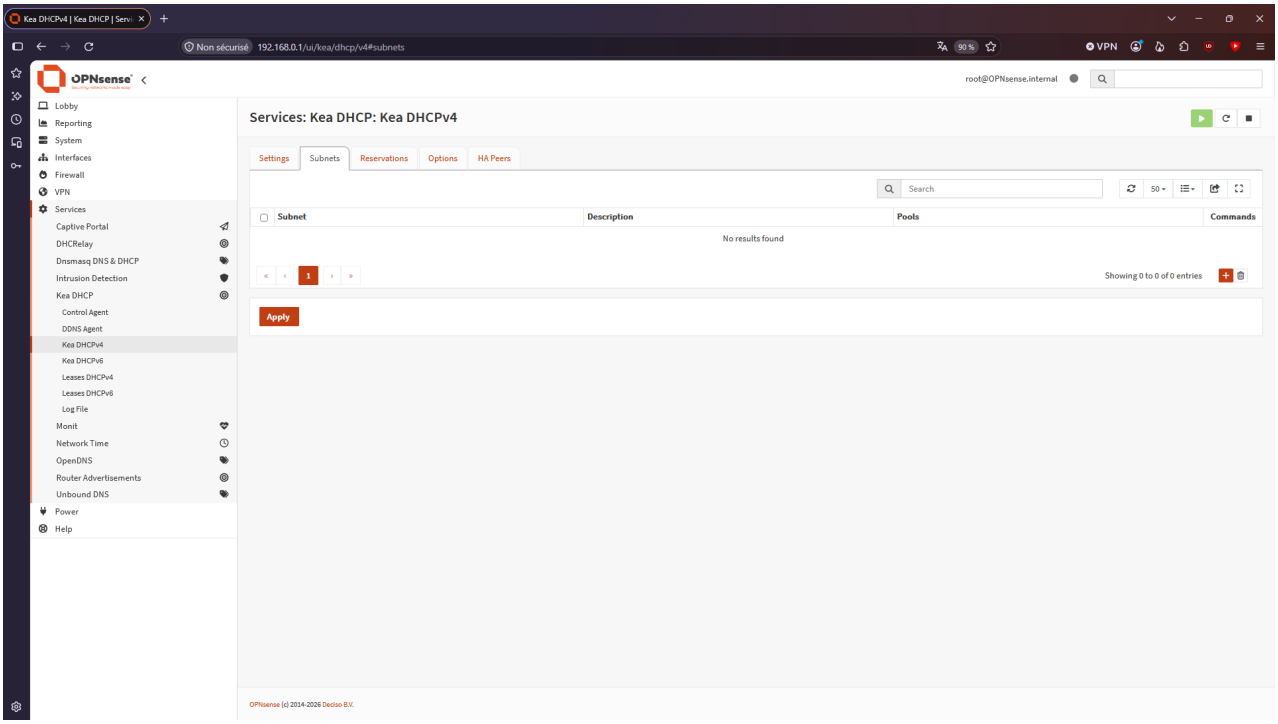
Vous allez procéder comme ceci :

1) Cochez **Enabled**

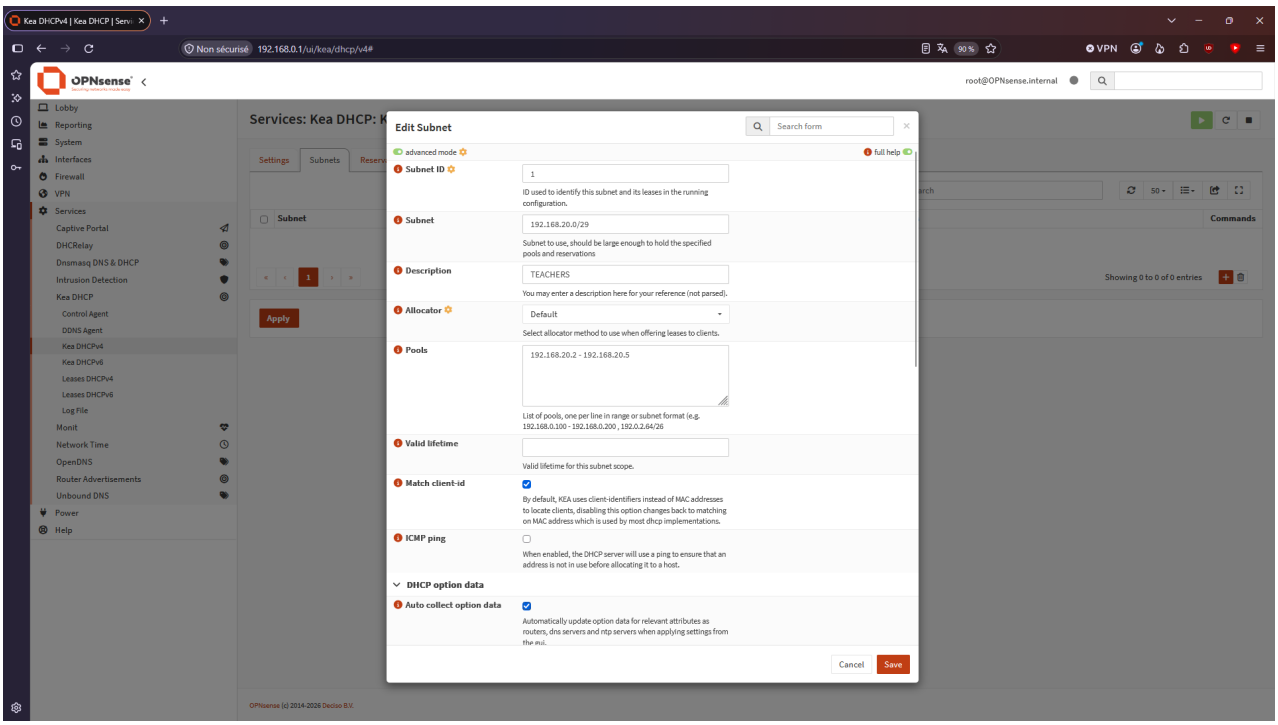
2) Sélectionnez les interface **20 (TEACHERS)** et **30 (STUDENTS)**
3) Activez **Firewall rules** pour autoriser les échanges DHCP
4) Appliquez

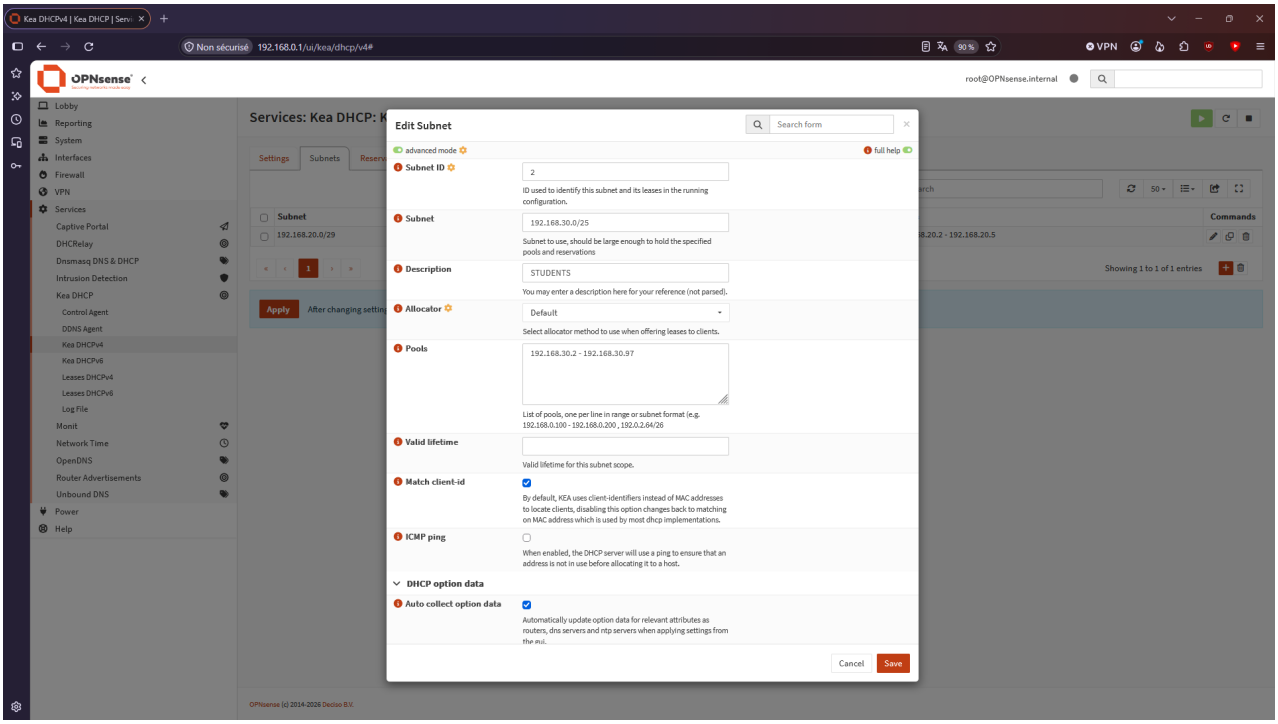


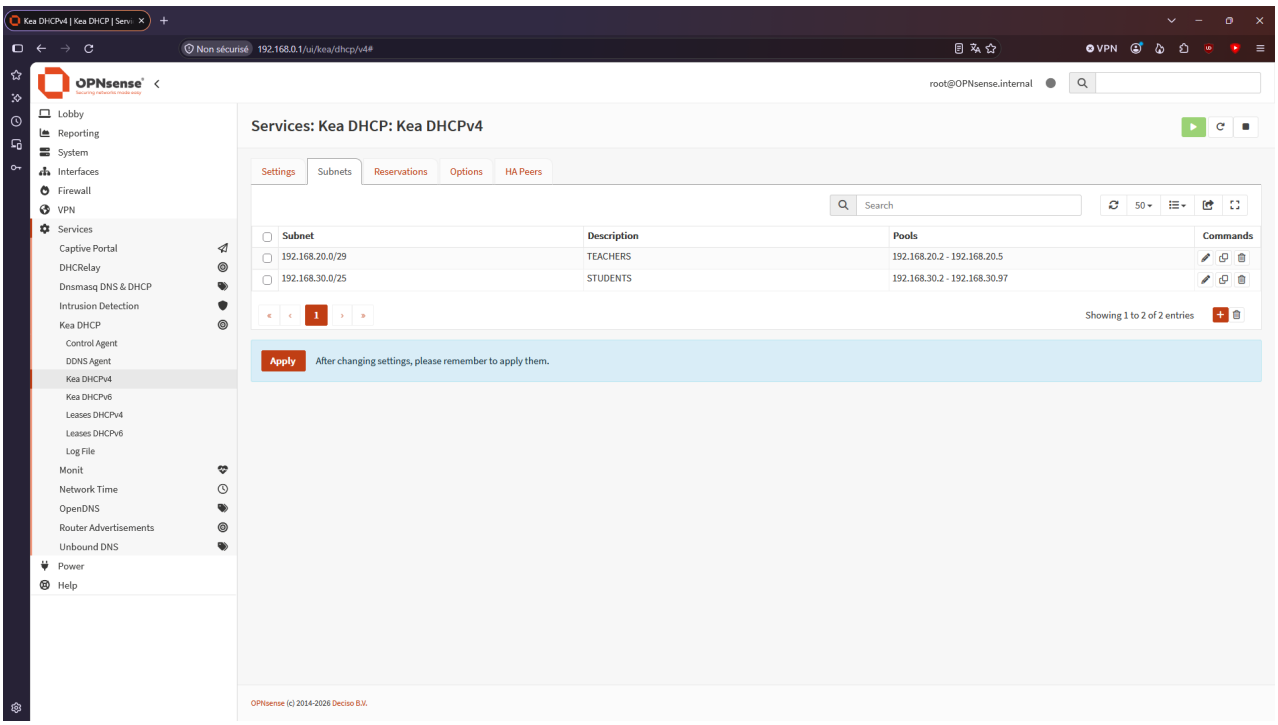
Allez ensuite dans **Subnets** :



Ajoutez les entrées suivantes pour chaque VLAN :


Edit Subnet
Search form
advanced mode
full help
Subnet ID
1
ID used to identify this subnet and its leases in the running configuration.
Subnet
192.168.20.0/29
Subnet to use, should be large enough to hold the specified pools and reservations
Description
TEACHERS
You may enter a description here for your reference (not parsed).
Allocator
Default
Select allocator method to use when offering leases to clients.
Pools
192.168.20.2 - 192.168.20.5
List of pools, one per line in range or subnet format (e.g. 192.168.0.100 - 192.168.0.200 , 192.0.2.64/26
Valid lifetime
Valid lifetime for this subnet scope.
Match client-id
By default, KEA uses client-identifiers instead of MAC addresses to locate clients, disabling this option changes back to matching on MAC address which is used by most dhcp implementations.
ICMP ping
When enabled, the DHCP server will use a ping to ensure that an address is not in use before allocating it to a host.
DHCP option data
Auto collect option data
Automatically update option data for relevant attributes as routers, dns servers and ntp servers when applying settings from
Cancel
Save


Edit Subnet
Search form
advanced mode
full help
Subnet ID
2
ID used to identify this subnet and its leases in the running configuration.
Subnet
192.168.30.0/25
Subnet to use, should be large enough to hold the specified pools and reservations
Description
STUDENTS
You may enter a description here for your reference (not parsed).
Allocator
Default
Select allocator method to use when offering leases to clients.
Pools
192.168.30.2 - 192.168.30.97
List of pools, one per line in range or subnet format (e.g. 192.168.0.100 - 192.168.0.200 , 192.0.2.64/26
Valid lifetime
Valid lifetime for this subnet scope.
Match client-id
By default, KEA uses client-identifiers instead of MAC addresses to locate clients, disabling this option changes back to matching on MAC address which is used by most dhcp implementations.
ICMP ping
When enabled, the DHCP server will use a ping to ensure that an address is not in use before allocating it to a host.
DHCP option data
Auto collect option data
Automatically update option data for relevant attributes as routers, dns servers and ntp servers when applying settings from
Cancel
Save



Appliquez les modifications et vous aurez vos serveurs DHCP pour les VLAN 20 et 30. Vous pouvez vérifiez le DHCP sur un poste client.

Création de Proxmox VE (REVOIR)

Proxmox VE est une plateforme open source qui permet d'exéctuer plusieurs systèmes isolés sur une machine physique. Nous allons l'utiliser pour faire de la virtualisation imbriquée en regroupant les machines de l'infrastructure.

1. Créer la VM

Nous allons mettre en place Proxmox VE comme VM de Hyper-V. Commencez par récupérer l'ISO de Proxmox VE version 8.4 à cette adresse https://www.proxmox.com/en/downloads/proxmox-virtual-environment. Puis exécutez le script suivant en respectant les paramètres :

```
# Configuration
$PVE = [PSCustomObject]@{
    Name           = "pve"
    Generation     = 2
    CPUCount       = 6
    VRAMBytes      = 20GB
    Path           = "V:\Kiformatou\pve"
    VHDPath        = "V:\Kiformatou\pve\pve.vhdx"
    VHDSizeBytes   = 300GB
    SwitchName     = "KIFORMATOU"
    ISOPath        = "V:\ISO\pve_8.4-1.iso"
}

# Create the VM directory
New-Item -ItemType Directory -Path $PVE.Path -Force | Out-Null

# Create a Generation 2 VM with 20 GB vRAM and a 300 GB vHD
New-VM -Name $PVE.Name `
    -Generation $PVE.Generation `
    -MemoryStartupBytes $PVE.VRAMBytes `
    -Path $PVE.Path `
    -NewVHDPath $PVE.VHDPath `
    -NewVHDSizeBytes $PVE.VHDSizeBytes `
    -SwitchName $PVE.SwitchName

# Assign 6 virtual processors
```

```
Set-VMProcessor -VMName $PVE.Name -Count $PVE.CPUCount

# Disable secure boot
Set-VMFirmware -VMName $PVE.Name -EnableSecureBoot Off

# Activate exposed virtualization
Set-VMProcessor -VMName $PVE.Name -ExposeVirtualizationExtensions $true

# Activate Bridge
Get-VMNetworkAdapter -VMName $PVE.Name |
    Set-VMNetworkAdapter -MacAddressSpoofing On

# Attach the installation ISO and boot from it
$dvdDrive = Add-VMDvdDrive -VMName $PVE.Name -Path $PVE.ISOPath -Passthru
Set-VMFirmware -VMName $PVE.Name -FirstBootDevice $dvdDrive

# Disable dynamic memory
Set-VMMemory -VMName $PVE.Name -DynamicMemoryEnabled $false -
StartupBytes $PVE.VRAMBytes

# Display the VM status
Get-VM -Name $PVE.Name
Get-VMProcessor -VMName $PVE.Name |
    Select-Object VMName, ExposeVirtualizationExtensions
```

Puis, démarrez :

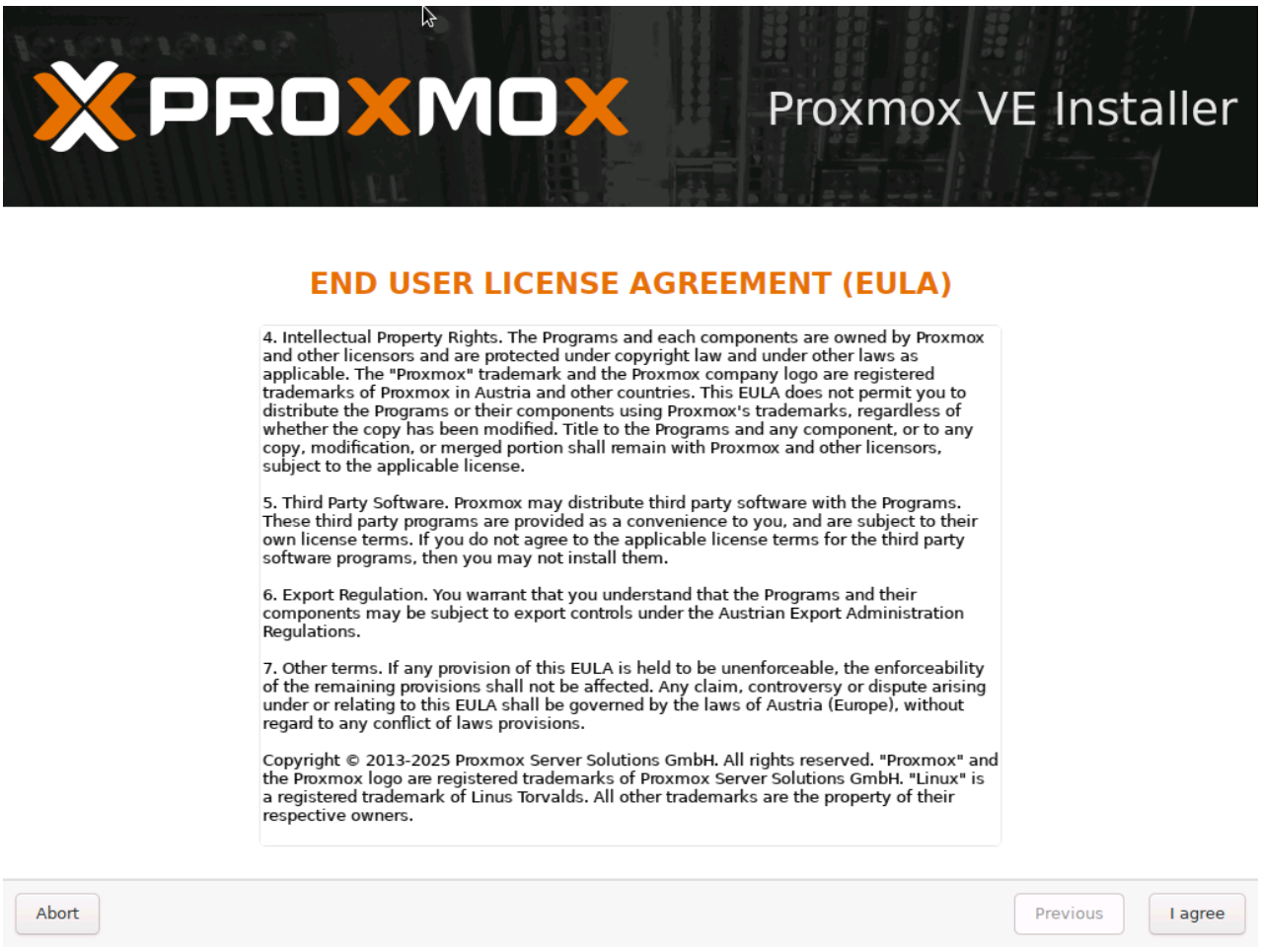
```
Start-VM -Name "pve"
vmconnect.exe localhost "pve"
```

2. Installer la VM

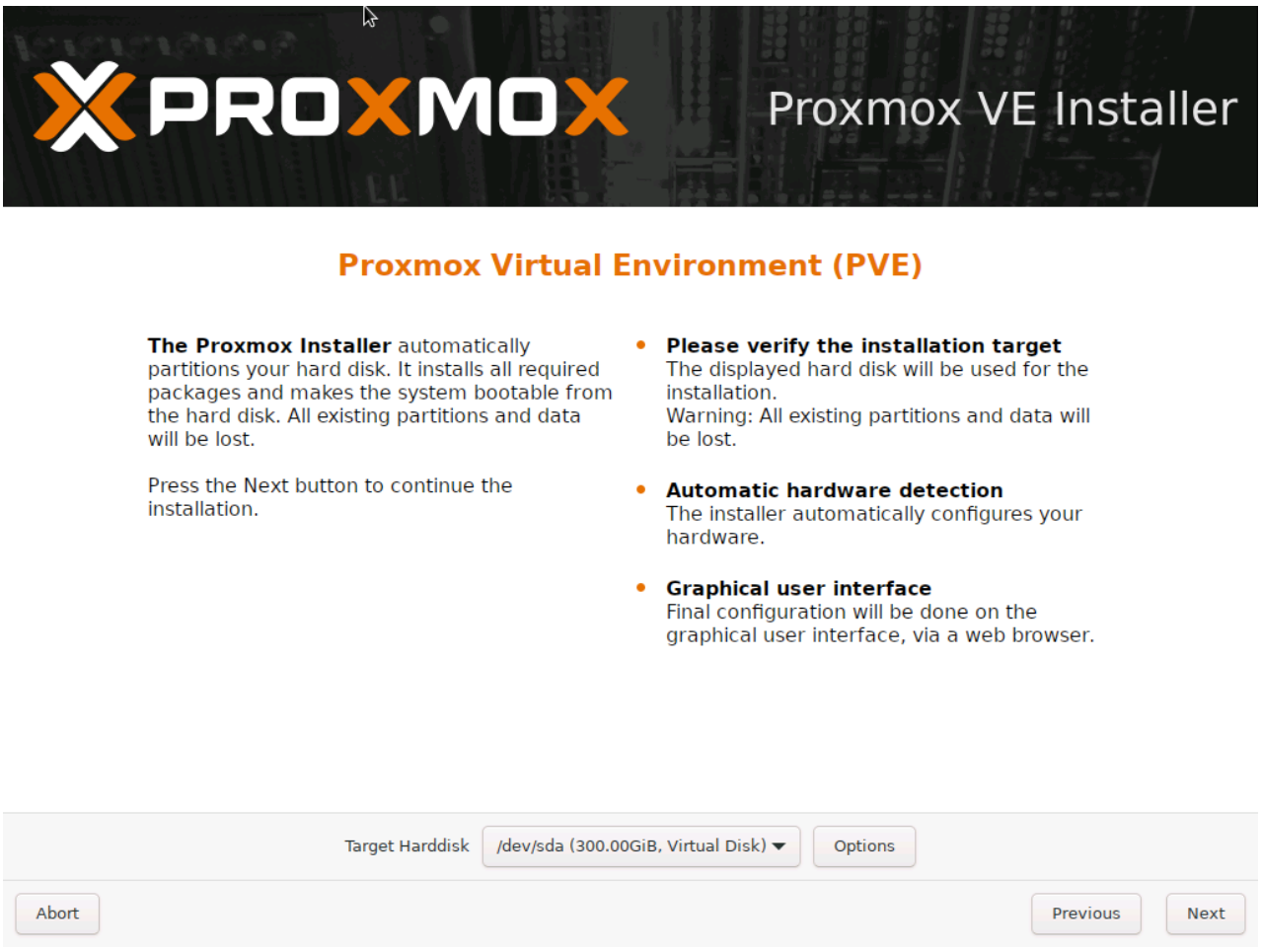
Après démarrage de la VM, choisissez **Install Proxmox VE (Graphical)** :



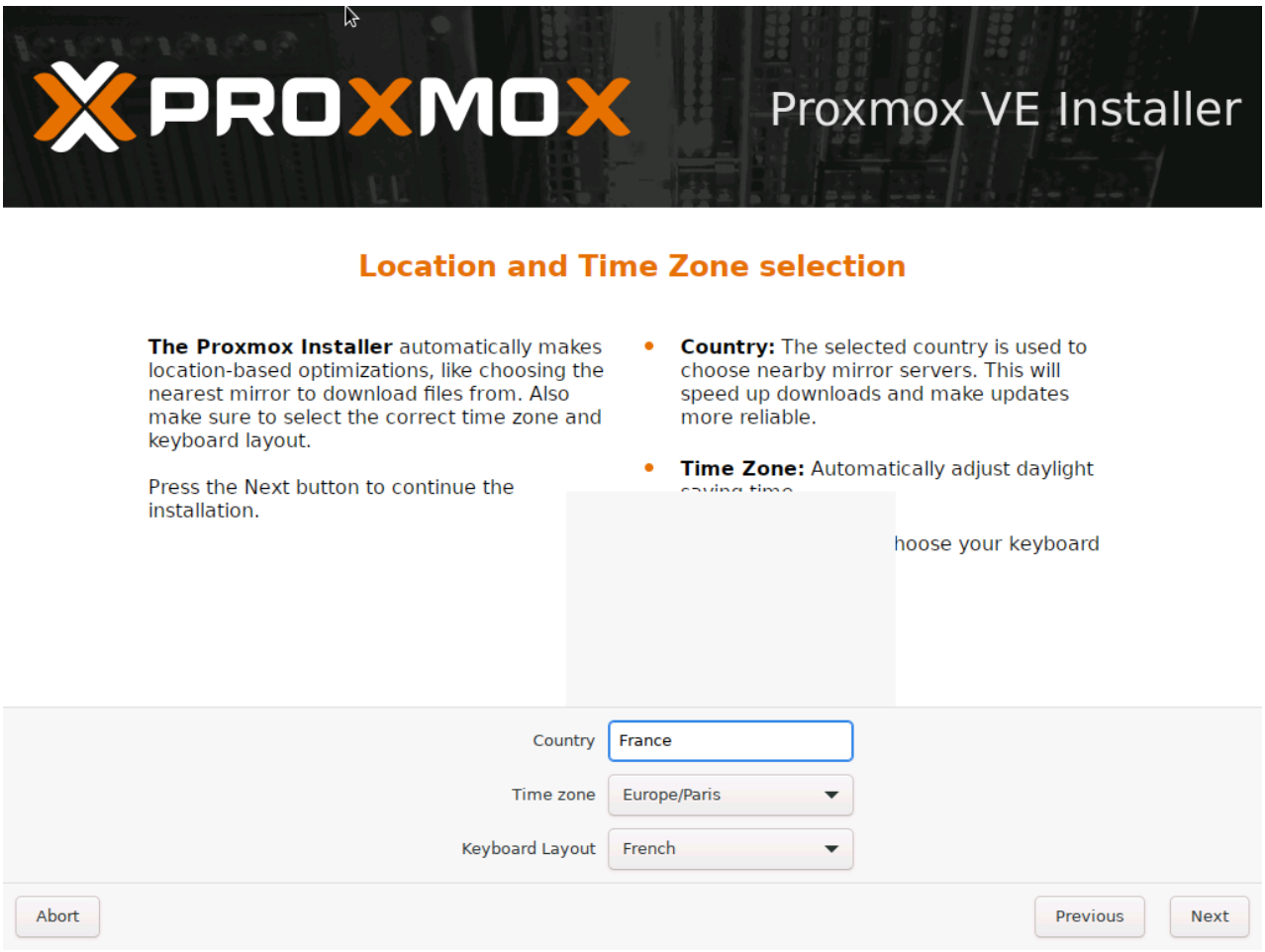
Acceptez **EULA :**



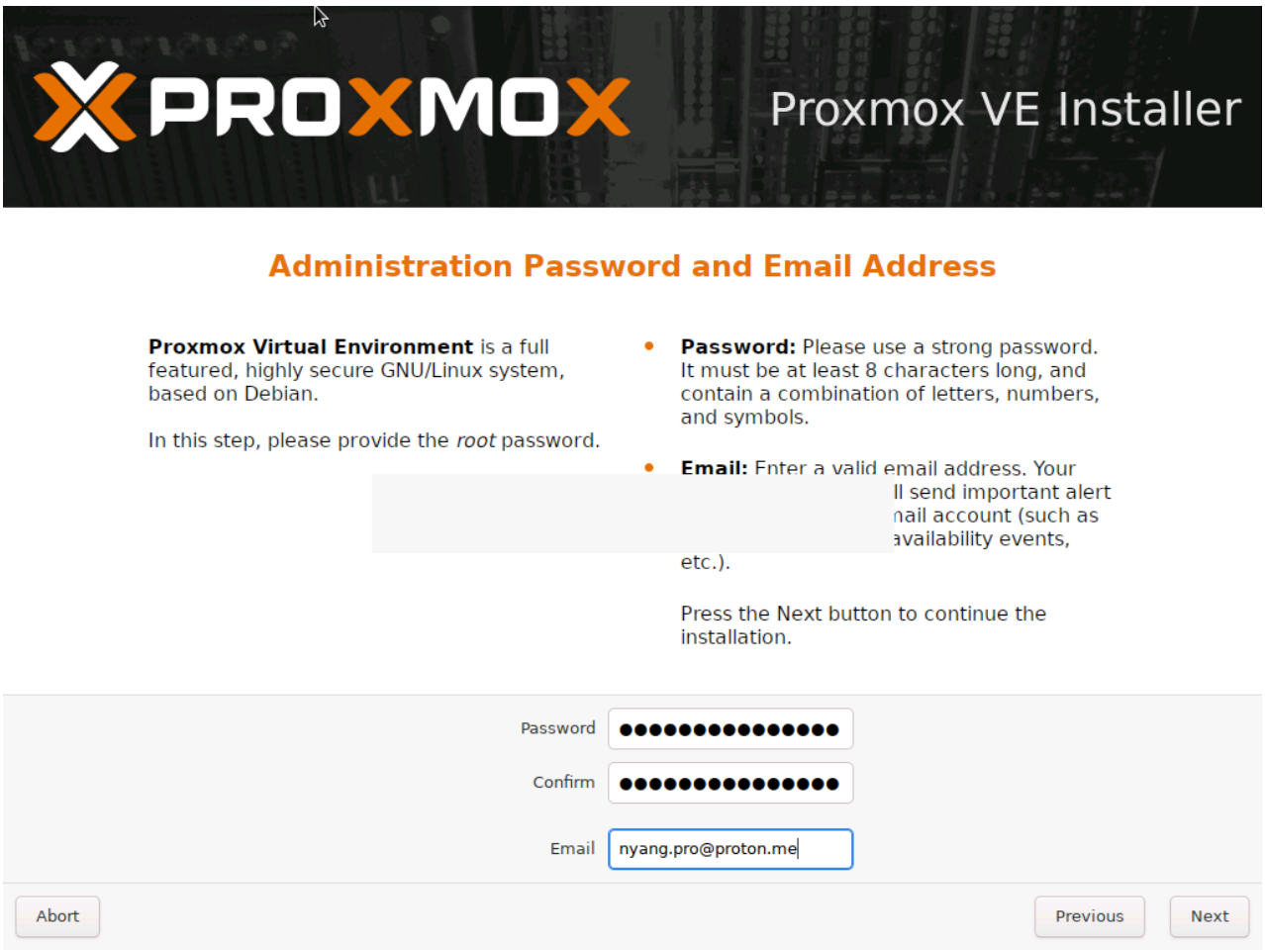
Cliquez sur **Next**



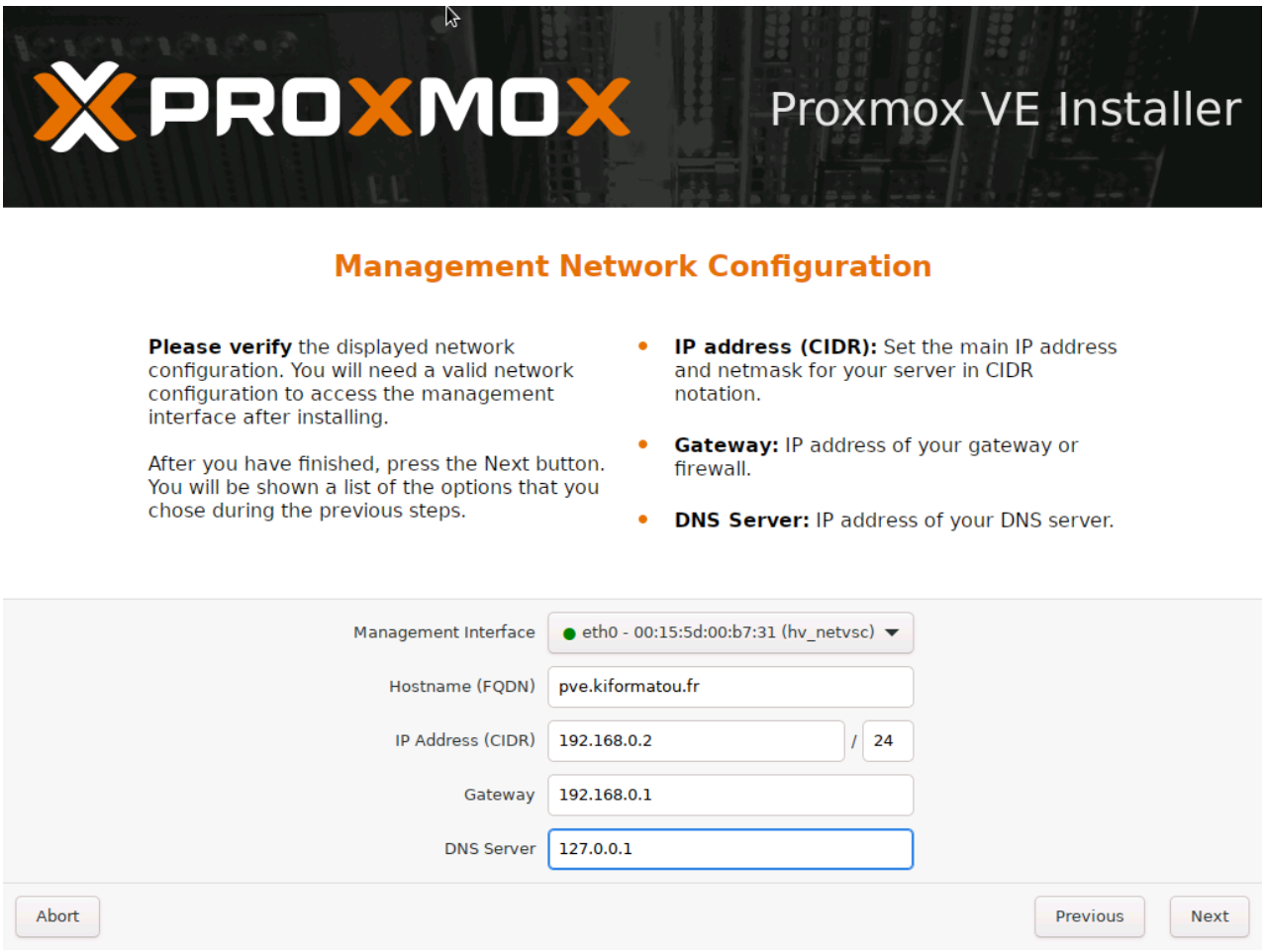
Tapez **France**, les paramètres restants se mettrons à jour automatiquement :



Tapez un mot de passe votre choix puis un email :



Entrez les paramètres suivants pour la configuration réseau :



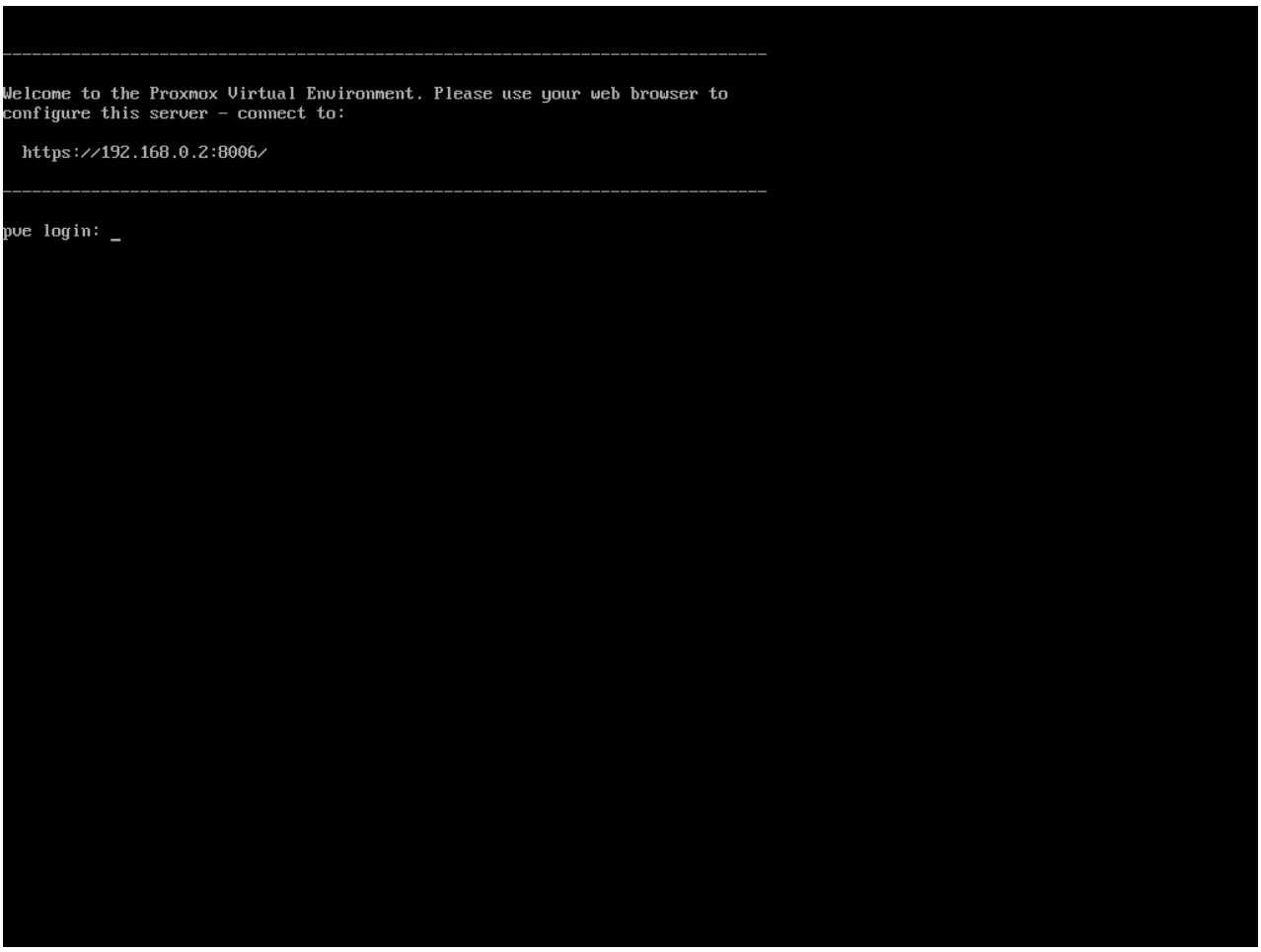
Vérifiez vos informations et débutez l'installation :



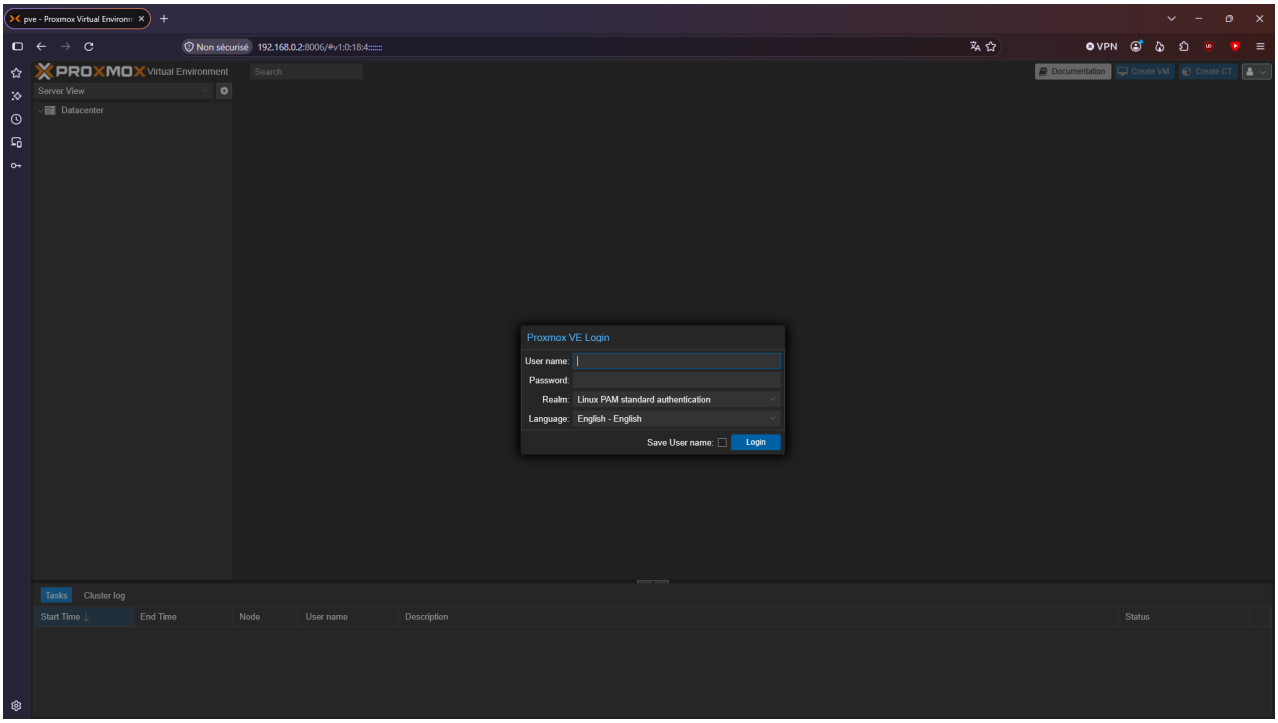
3. Configurer la VM

3.1. Mettre à jour Proxmox

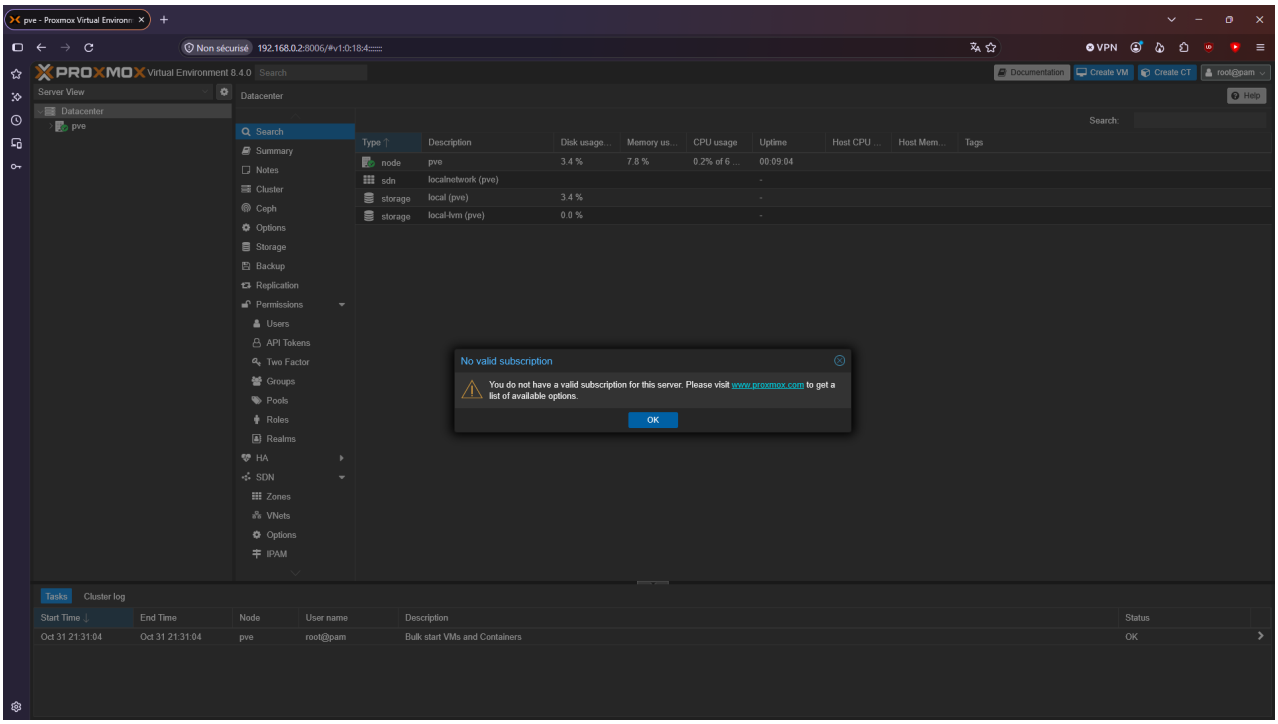
Après avoir terminé l'installation, la VM redémarrera. Vous devez tomber sur l'affichage suivant :



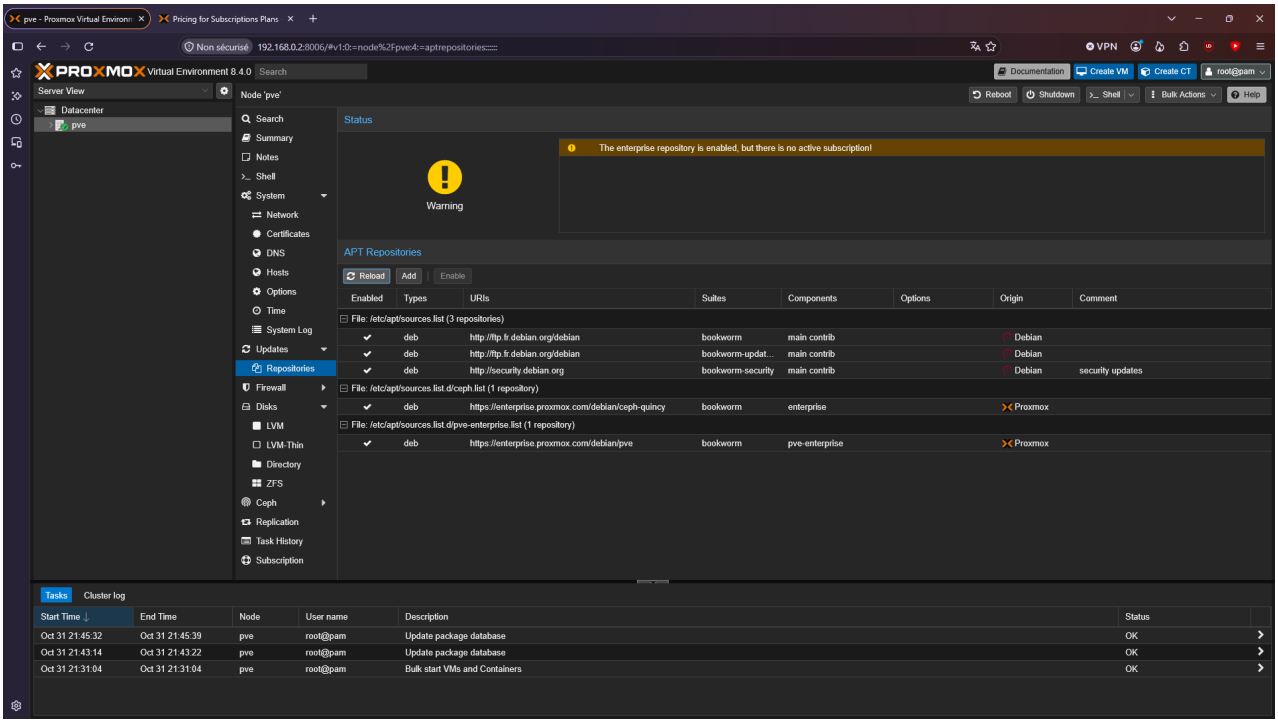
Accéder à l'interface Web de Proxmox VE en allant à l'adresse https://192.168.0.2:8006 :



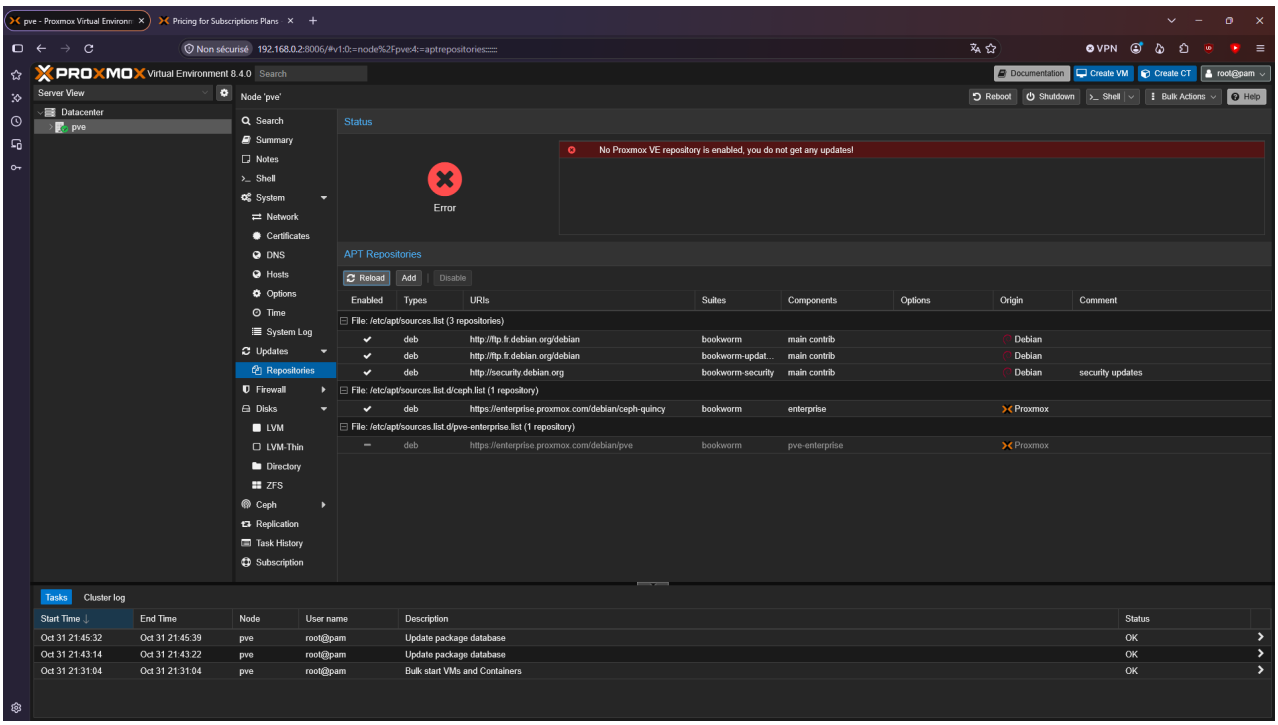
Entre vos identifiants pour **root**, vous serez ensuite sur le tableau de bord de Proxmox :



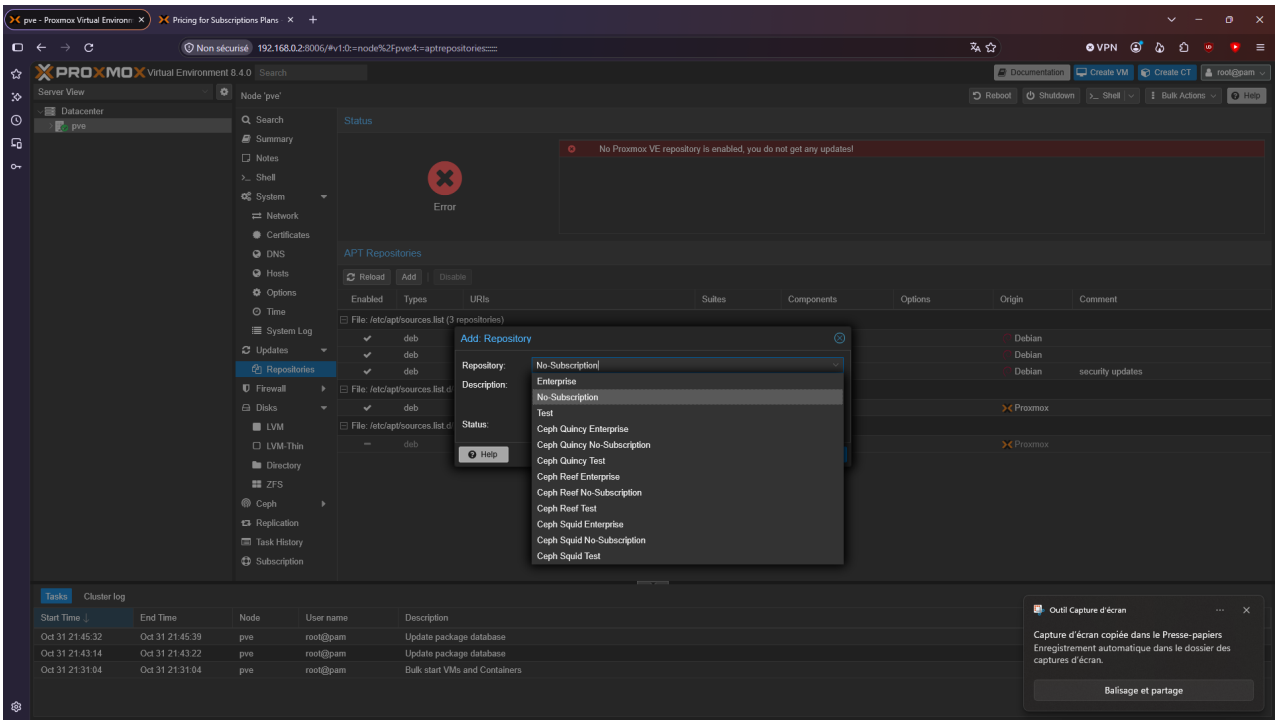
Commencez par mettre à jour Proxmox en cliquant sur votre noeud **pve -> Updates -> Repositories** :



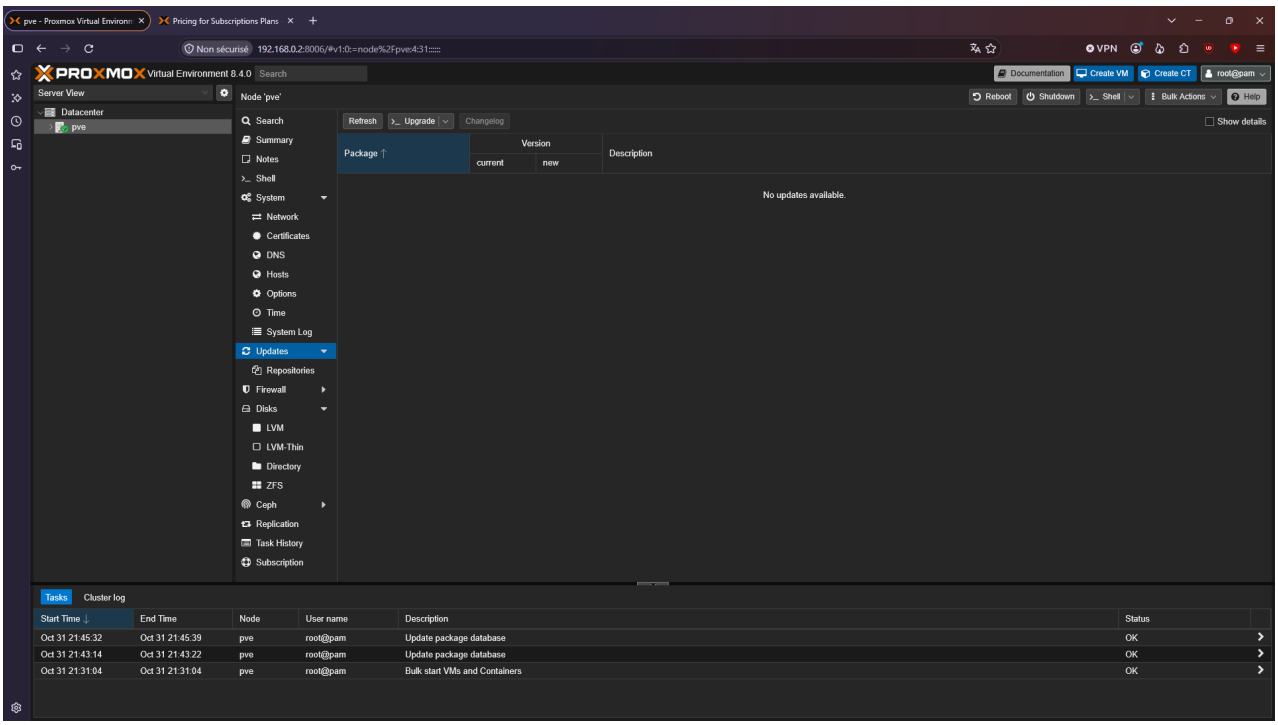
Désactivez les composants **enterprise** et **pve-enterprise** :



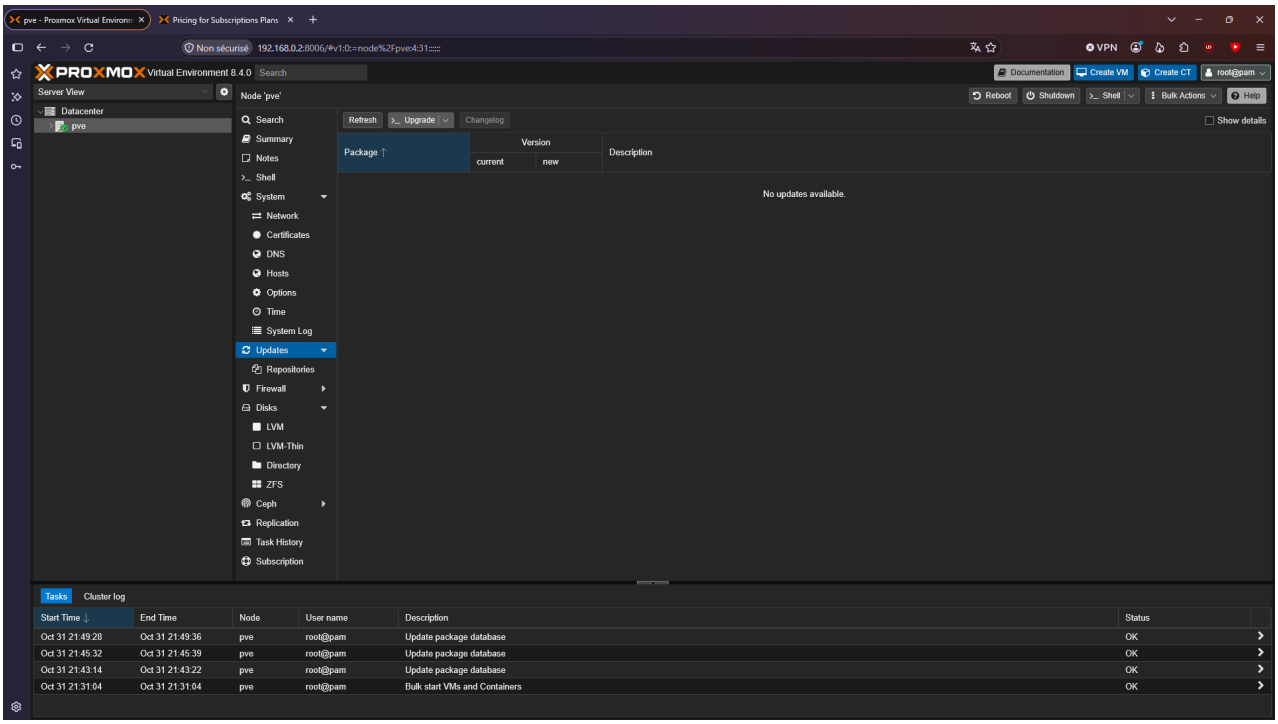
Cliquez sur **Add**, puis choisissez **No-Subscription** :

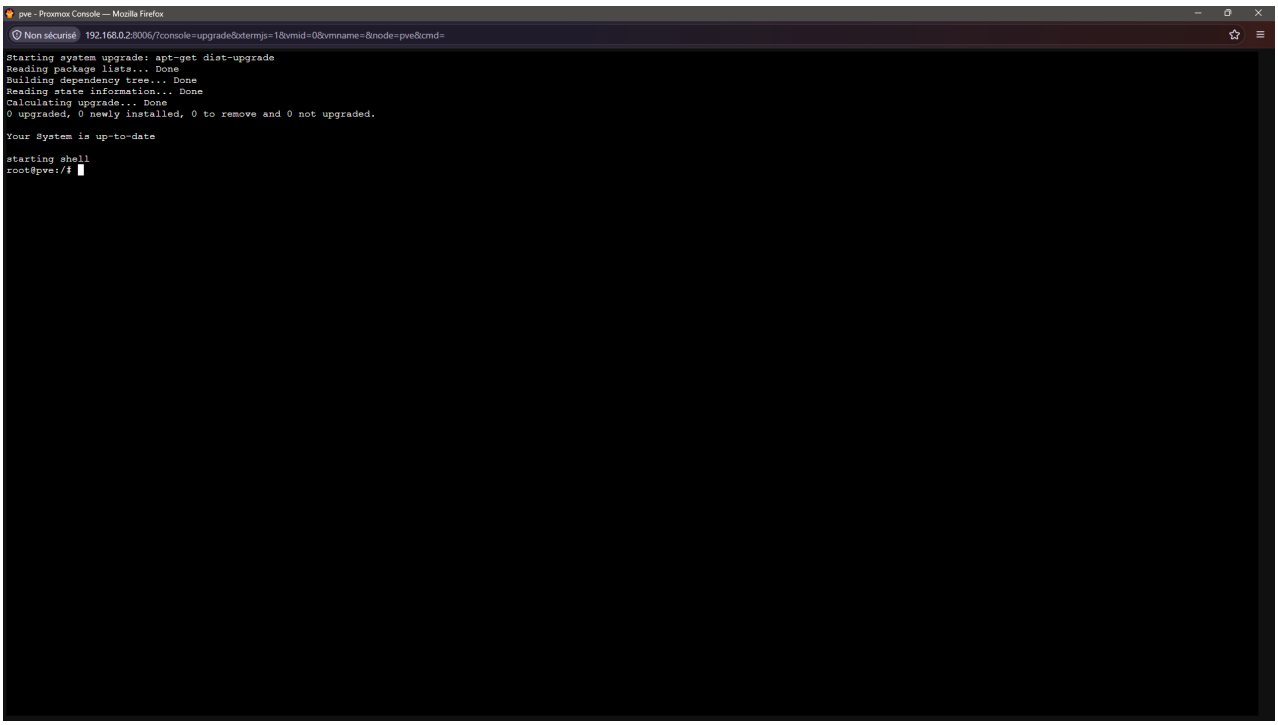


Allez ensuite dans **Updates** puis cliquez sur **Refresh** :



Puis cliquez sur **Update** :

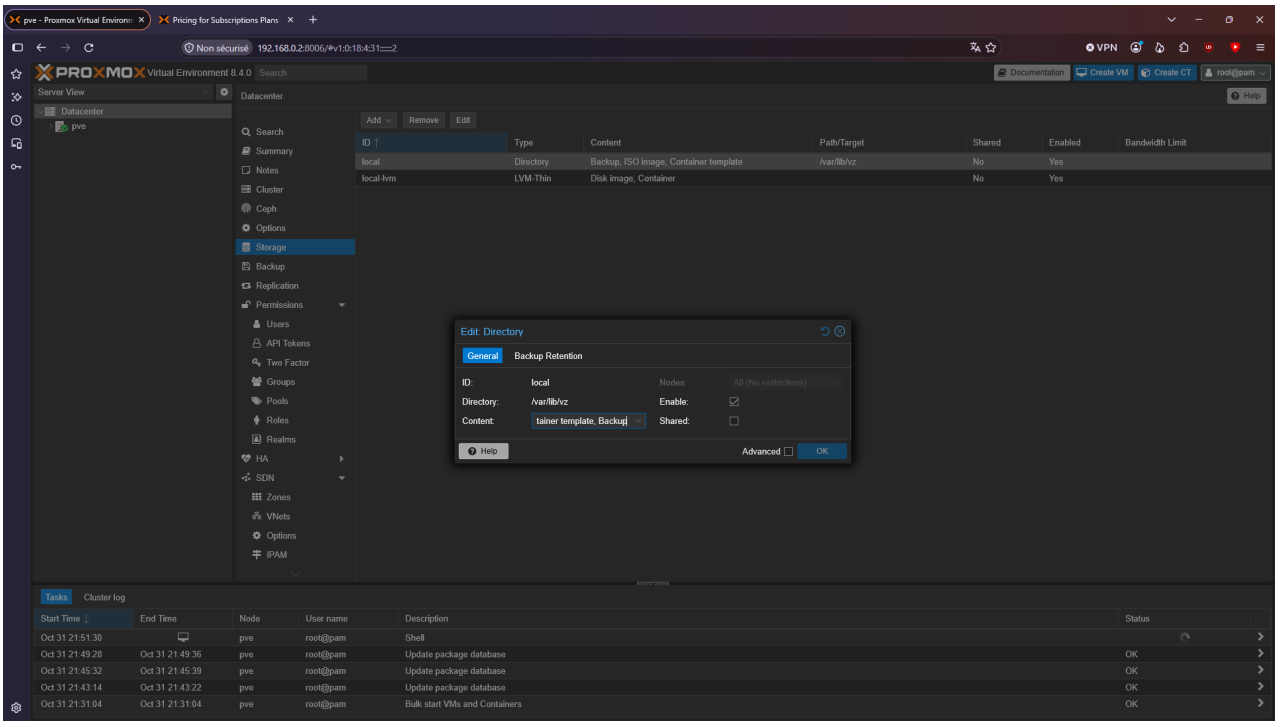




Nous avons ainsi installer et mis à jour Proxmox VE, pensez à exporter la VM comme Template du projet.

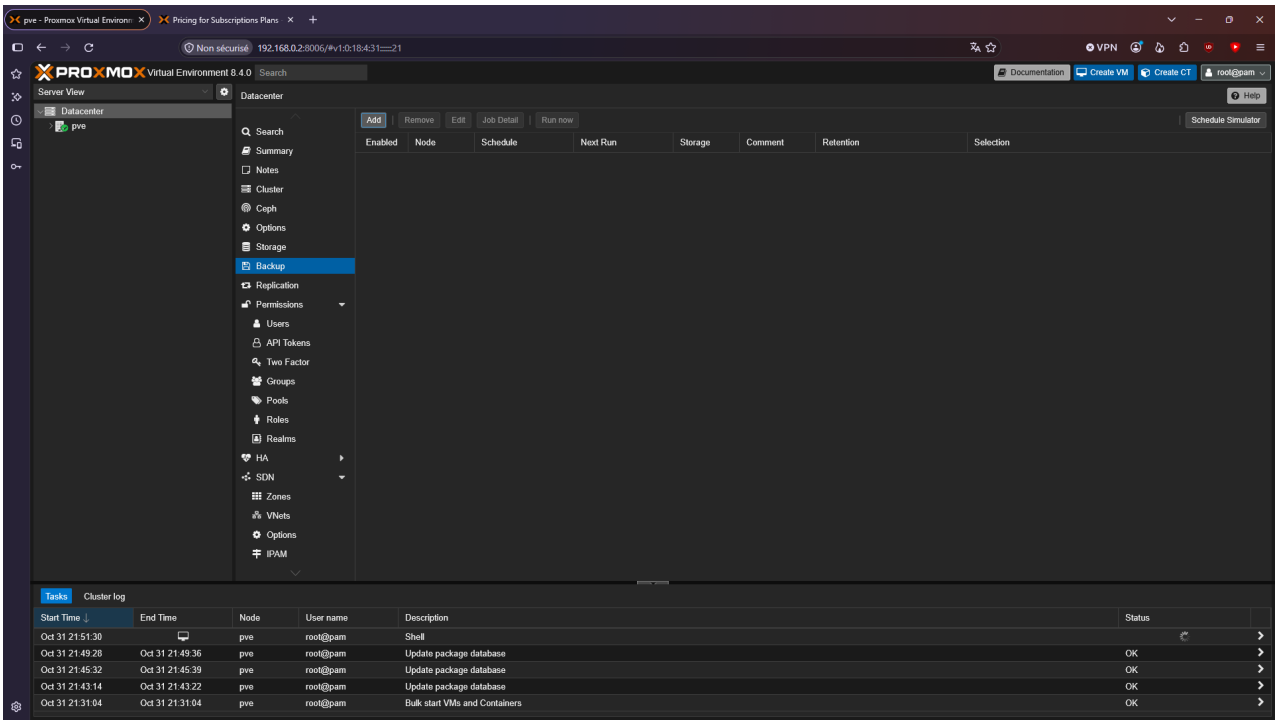
3.2. Configurer les sauvegardes de secours sur Proxmox

Nous allons maintenant configurer les sauvegardes des VM sur le disque virtuel interne de Proxmox. Pour cela, allez dans **Datacenter -> Storage -> local -> Edit** :

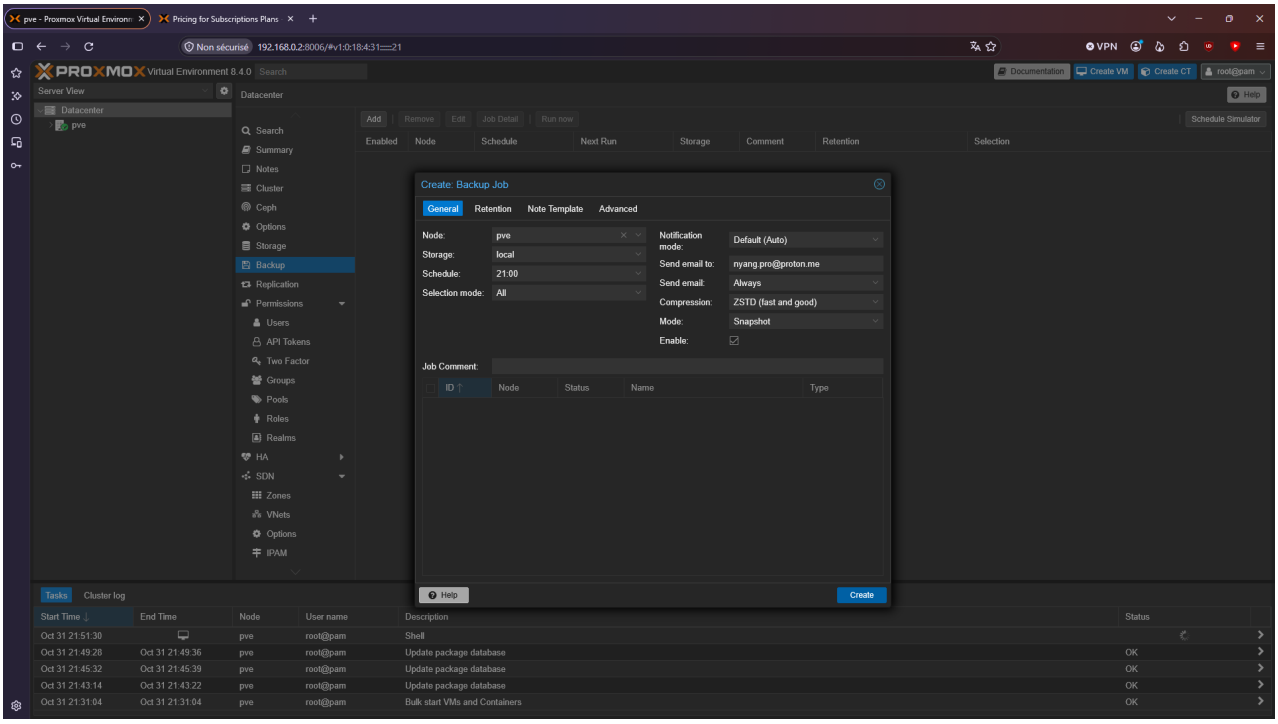


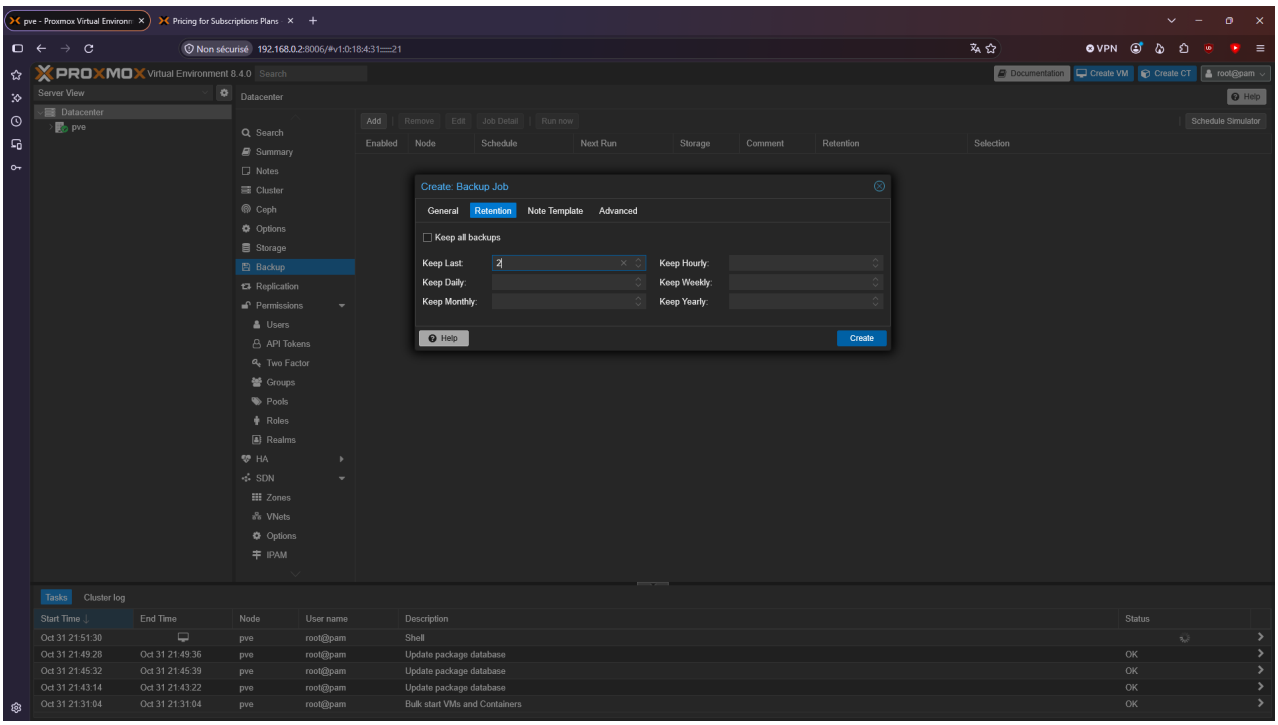
Assurez-vous que **Content** contient **Backup** avant de procéder à l'étape suivante.

Ouvrez **Datacenter -> Backup -> Add** :

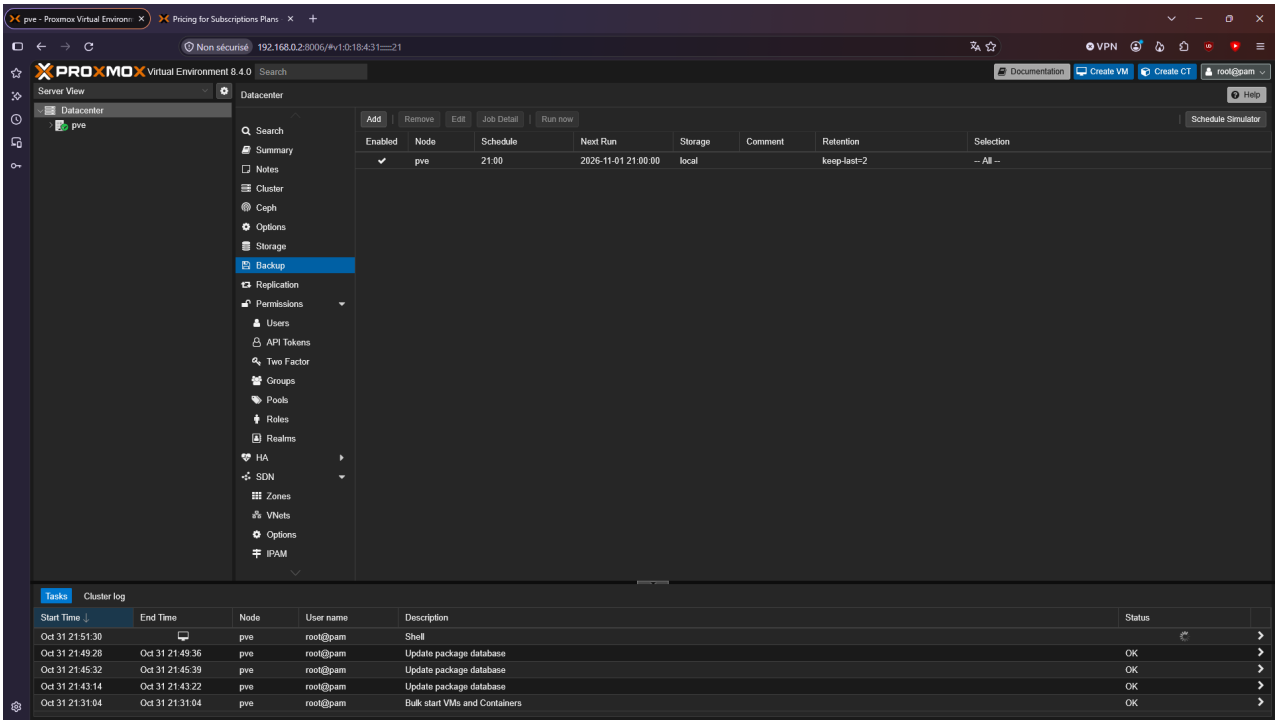


Ajoutez les paramètres suivants :





Puis créez la sauvegarde de secours.



Vous avez donc mis en place une sauvegarde de secours pour vos VM automatiquement.

Création des postes de travail Windows 10 (WIP)

Création du serveur Active Directory et DNS (WIP)

Création du serveur Nextcloud (WIP)

Création du serveur de sauvegarde (WIP)

6. Support et références

6.1. Diagnostic et support

Journaux [Emplacement ; accès ; rétention ; requête utile.]

Symptôme	Vérification	Action ou escalade
[Accès impossible]	[Réseau ; identité ; état du service]	[Action documentée ; contact]
[Erreur ou lenteur]	[Journaux ; ressources ; dépendances]	[Action documentée ; contact]

Support [Équipe ; canal ; horaires ; informations à fournir au ticket.]

6.2. Glossaire

- **VM** : Machine Virtuelle (Virtual Machine)
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6.3. Références

Documentations

- Microsoft. Documentation de Hyper-V. 2026.

- Microsoft. Get-VMSwitch. 2026.
- Deciso. Virtual & Cloud-Based Installation. 2026.
- Microsoft. Configuration requise pour Windows 10. 2026.
- Microsoft. Configuration matétrielle requise pour Windows Server. 2026.

Ressources externes

- Installation d'OPNsense